

Choosing What to Calibrate and What to Estimate in Structural Models

JOAN ALEGRE CANTON

Department of Economics, Universidad Carlos III de Madrid

Structural models often fix (calibrate) some parameters and estimate the rest, but this calibration–estimation partition is usually chosen by convention. This paper treats that choice as an econometric partition-selection problem. For each admissible partition, we construct a scalar sensitivity statistic measuring the local response of a target object—such as a policy effect, welfare measure, impulse response, or treatment effect—to perturbations of the calibrated parameters. The selected partition minimizes this statistic and therefore minimizes worst-case local bias from calibration errors. We first illustrate the decision problem in two canonical examples. We then apply it to the New Keynesian model of Nakamura and Steinsson (2018), where the partition choice has large implications for credibility: some partitions remain reliable under sizeable miscalibrations, whereas others generate large bias from small calibration errors. The procedure requires only local derivatives, avoids repeated re-estimation, and applies to a broad class of structural models.

KEYWORDS. Model selection, Structural parameters, Sensitivity analysis, Partial identification.

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Joan Alegre Canton: joan.alegre.canton@gmail.com

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1. INTRODUCTION

A common practice in structural model estimation is to calibrate some structural parameters—that is, to hold them fixed at prespecified values—and estimate the rest.¹ This approach is often justified when full estimation is computationally infeasible (Jørgensen, 2023), when the model is not fully identified (Iskrev, 2019), or when population values are known. In practice, however, the calibration–estimation partition is rarely chosen by a formal econometric criterion. It is often inherited from previous studies, justified informally, or dictated by computational convenience.

This paper provides an econometric criterion for choosing what to calibrate and what to estimate in a general class of structural models. We treat the calibration–estimation split as a partition-selection problem. For each admissible partition, we construct a scalar sensitivity statistic that measures the local response of the object of interest to perturbations of the calibrated parameters. We then select the admissible partition that minimizes this sensitivity. The resulting rule chooses the partition that delivers the smallest worst-case local bias induced by calibration error. This perspective connects the paper to robust estimation under local misspecification, especially Bonhomme and Weidner (2022).

A natural alternative would be to select the partition by minimizing an MSE criterion that trades off calibration bias and sampling variance. We focus instead on bias, through the sensitivity statistic, because implementing an MSE criterion requires the full joint variance–covariance matrix of the stacked moments or reduced-form estimates. This requirement is often restrictive when different moments are estimated from different datasets and not all cross-covariances are identified.

¹Throughout the paper, a parameter is *calibrated* if it is held fixed during the estimation problem. This usage is narrower than in some applied and econometric work, where *calibration* refers to choosing parameters to match empirical moments, i.e. to minimum-distance estimation; see, for example, Cocci and Plagborg-Møller (2021). In our terminology, the latter is estimation.

A key feature of the implementation is that the least-sensitive partition is computed relative to two inputs. The first is a reference point for the structural parameters, typically obtained from a first-stage estimation. This reference point is where local sensitivities are evaluated. It also allows the researcher to compare candidate partitions without re-estimating the model separately for each one. The second input is a normalization of the calibrated parameters. This normalization specifies the scale on which calibration errors are measured and makes perturbations comparable across parameters expressed in different units. Hence, the least-sensitive partition should be interpreted as conditional on both the reference point and the chosen normalization of calibration errors.

The method is developed in a minimum-distance framework ([Newey and McFadden, 1994](#)), broad enough to cover several estimation approaches commonly used in structural work, including the Generalized Method of Moments (GMM), Classical Minimum Distance (CMD), Simulated Method of Moments (SMM), or Indirect Inference (II).

We illustrate the importance of partition choice in three settings. The first two are New Keynesian models, based on [An and Schorfheide \(2007\)](#) and [Nakamura and Steinsson \(2018\)](#). The third is the dynamic discrete-choice model in [Kalouptsi et al. \(2021\)](#). Across these examples, we show that the calibration-estimation partition has large implications for robustness. In the Nakamura-Steinsson application (Section 6), where the object of interest is the monetary policy's information-effect parameter, the sensitivity statistic varies substantially across admissible partitions. For some partitions, miscalibrations of only a few percent are sufficient to overturn the qualitative conclusion about the object of interest. By contrast, miscalibrations of comparable magnitude generate only a small worst-case bias under the least-sensitive partition. These results show that fragility can be avoided purely by choosing more carefully which parameters to fix and which to estimate.

The paper is related to several strands of literature. It builds most directly on work on sensitivity analysis for structural models. [Iskrev \(2019\)](#) and [Jørgensen](#)

(2023) study how estimates and model outputs respond to perturbations in calibrated parameters, while [Lau \(2024\)](#) develops local and global sensitivity measures for dynamic discrete choice models with fixed parameters. Relative to [Jørgensen \(2023\)](#), our proposed statistic is scalar and independent of calibrated parameter units. Related work studies sensitivity to moments or identifying assumptions, including [Andrews et al. \(2017\)](#), [Honore et al. \(2020\)](#), and [Escanciano and Zhu \(2013\)](#). Our contribution differs in that sensitivity is not the final object of interest. Instead, it is the input into a decision rule that selects which parameters should be fixed and which should be estimated.

The paper is also related to the literature on identification and partial identification of counterfactuals in dynamic discrete choice models. Building on [Aguirregabiria \(2010\)](#), [Aguirregabiria and Suzuki \(2014\)](#), [Norets and Tang \(2014\)](#), and [Arcidiacono and Miller \(2020\)](#), [Kalouptside et al. \(2021\)](#) characterize which counterfactuals are point identified despite payoff nonidentification, while [Kalouptside et al. \(2026\)](#) characterize the corresponding identified set for the counterfactual. We reverse the direction of this exercise. In that literature, the calibration or identifying strategy is taken as given, and the question is which counterfactuals are point or set identified. Here, the counterfactual exercise is taken as given, and the question is which normalization—or calibration—estimation partition—makes that same object of interest least-sensitive to local calibration error.

The minimax interpretation connects the paper to the literature on estimation and inference under local misspecification (e.g., [Newey, 1985](#), [Conley et al., 2012](#), [Kitamura et al., 2013](#)). The closest paper is [Bonhomme and Weidner \(2022\)](#), who choose an estimator, through its influence function, to minimize worst-case mean squared error under local misspecification. We differ in two main respects: our criterion targets worst-case bias rather than worst-case MSE, and our decision variable is the calibration–estimation partition rather than the influence function. Also closely related is [Armstrong and Kolesár \(2021\)](#), who show that GMM weights should account for both sampling variance and misspecification bias when moments may be only approximately valid. We apply a similar local-robustness logic to a different margin: the choice of which parameters to calibrate and which to estimate.

The remainder of the paper is organized as follows. Section 2 presents two motivational examples. Section 3 defines the framework and discusses the construction of the partition-selection methodology in detail. Section 4 offers a risk-minimization interpretation and relates the methodology to Bonhomme and Weidner (2022). Section 5 provides a step-by-step implementation guide. Section 6 applies the method to Nakamura and Steinsson (2018), and Section 7 presents Monte Carlo evidence.

2. TWO MOTIVATIONAL EXAMPLES

This section illustrates the importance of choosing the calibration–estimation partition. In both examples, the model requires fixing at least one structural parameter before the remaining parameters can be estimated. The central point is that different admissible choices of the fixed parameter can generate very different sensitivity of the object of interest to calibration error.

2.1 *A dynamic discrete choice model*

Consider the entry–exit model in Kalouptsi et al. (2021). In each period, a firm chooses whether to be inactive, $a = 0$, or active, $a = 1$. The state is $x = (k, w)$, where $k \in \{0, 1\}$ is the firm’s lagged activity status and $w \in \{w_L, w_M, w_H\}$ is an exogenous demand state. Up to the idiosyncratic choice shock, flow payoffs are

$$\pi_a(k, w; \eta) = \begin{cases} k \, sv, & a = 0, \\ k\{\bar{\pi}(w) - fc\} - (1 - k)ec, & a = 1, \end{cases} \quad \eta = (ec, fc, sv).$$

Here ec is the entry cost, fc is the fixed cost, and sv is the scrap value. Given η , the choice-specific value and CCP are

$$v_a(x; \eta) = \pi_a(x; \eta) + \rho \mathbb{E}[V(x'; \eta) \mid a, x],$$

$$p_a(x; \eta) = \frac{\exp\{v_a(x; \eta)\}}{\exp\{v_0(x; \eta)\} + \exp\{v_1(x; \eta)\}},$$

where the second expression uses the type-I extreme-value normalization.²

The counterfactual policy is a 20% proportional tax on the active operating payoff. Let $\tau = 0.20$. The counterfactual flow payoff is

$$\pi_0^\tau(k, w; \eta) = \pi_0(k, w; \eta), \quad \pi_1^\tau(k, w; \eta) = (1 - \tau) k \{ \bar{\pi}(w) - fc \} - (1 - k)ec.$$

The object of interest is the counterfactual CCP vector,

$$p_1^\tau(x; \eta).$$

The model is not point identified without one additional normalization: one of ec , fc , or sv must be fixed (Aguirregabiria and Suzuki, 2014). This is not merely a computational detail. Observationally equivalent parameterizations of the model can imply different values of $p_1^\tau(x; \eta)$. In applications, researchers most often fix either the scrap value or the fixed cost. Table 1 summarizes this choice in related studies.

²The numerical primitives are those in Kalouptsi et al. (2021). Writing ξ for the inverse-demand slope, residual demand is $P_t = w_t - \xi Q_t$, marginal cost is $c = 11$, $\xi = 1.5$, $w = (20, 17, 12)$, and $\bar{\pi}(w) = (w - c)^2 / (4\xi)$. The remaining primitives are $\rho = 0.95$, $ec = 9$, $fc = 5.5$, $sv = 10$, and

$$F_w = \begin{pmatrix} 0.40 & 0.35 & 0.25 \\ 0.30 & 0.40 & 0.30 \\ 0.20 & 0.20 & 0.60 \end{pmatrix}.$$

The idiosyncratic shocks are type-I extreme value.

TABLE 1. Calibration Strategy in Recent Studies

<i>Calibrate the scrap value</i>	<i>Calibrate the fixed cost</i>
Das et al. (2007)	Ryan (2012)
Kryukov (2010)	Sweeting (2013)
Collard-Wexler (2011)	Santos (2017)
Collard-Wexler (2013)	Beresteanu et al. (2024)
Aguirregabiria and Ho (2012)	
Suzuki (2013)	
Barwick and Pathak (2015)	
Lin (2015)	
Igami (2017)	
Varela (2018)	

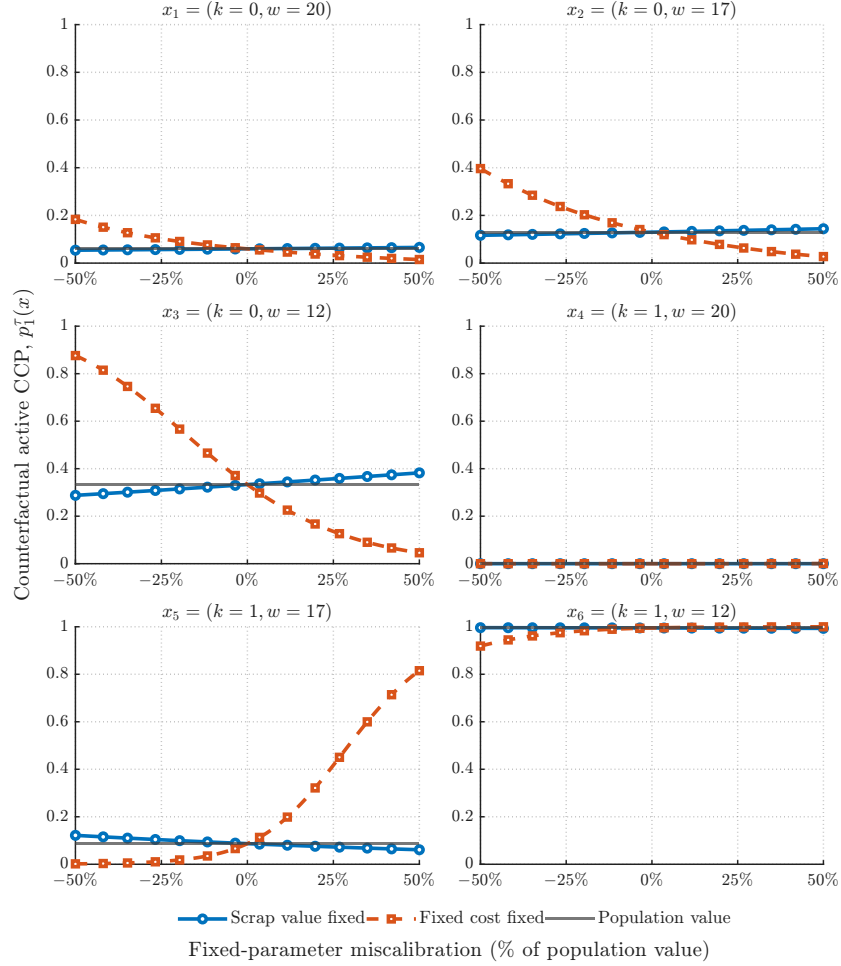
Note: No study in this list calibrated the entry cost.

Figure 1 plots the object of interest as a function of the fixed parameter. For each curve, either sv or fc is fixed at a value around its population value, the remaining cost parameters are then estimated, and the counterfactual CCP is re-computed. The horizontal axis is centered at the true value of the fixed parameter, so zero corresponds to correct calibration. The horizontal line is the population counterfactual CCP; the vertical distance from this line is the bias induced by miscalibration.

The figure shows that fixing the scrap value produces a relatively flat curve, while fixing the fixed cost produces a much steeper one. Thus, two common ways of identifying the same entry–exit model imply very different robustness properties for the same counterfactual object.

This motivational example is related to Kalouptside et al. (2021) and Kalouptside et al. (2026). Those papers take the calibration strategy as given and ask which counterfactuals are point identified, or what set of counterfactual values is implied when point identification fails. We ask the dual question. We take the counterfactual object, $p_1^T(x; \eta)$, as given and compare the admissible calibration

FIGURE 1. Counterfactual CCP as a function of the fixed parameter



strategies themselves. In this example, fixing sv, fc, or ec generates a different local range for the same counterfactual CCP under calibration errors; our method selects the strategy that makes this local range smallest.

2.2 A canonical New Keynesian model

The same issue arises in a standard macroeconomic environment. Consider the three-equation forward-looking New Keynesian model used in [An and Schorfheide \(2007\)](#):

Euler Equation:
$$y_t = \mathbb{E}_t y_{t+1} + g_t - \mathbb{E}_t g_{t+1} - \frac{1}{\tau} (r_t - \mathbb{E}_t \pi_{t+1} - \mathbb{E}_t z_{t+1}).$$

Phillips curve: $\pi_t = \rho \mathbb{E}_t \pi_{t+1} + \kappa(y_t - g_t),$

$$c_t = y_t - g_t.$$

Monetary Policy: $r_t = \rho_r r_{t-1} + (1 - \rho_r) \psi_1 \pi_t + (1 - \rho_r) \psi_2 (y_t - g_t) + \epsilon_{rt},$

$$g_t = \rho_g g_{t-1} + \epsilon_{gt},$$

$$z_t = \rho_z z_{t-1} + \epsilon_{zt}.$$

The shocks $\epsilon_{r,t}$, $\epsilon_{g,t}$, and $\epsilon_{z,t}$ are mutually uncorrelated, with variances σ_r^2 , σ_g^2 , and σ_z^2 ³.

Let the object of interest be ψ_1 , the coefficient on inflation in the Taylor rule. In this model, ψ_1 cannot be jointly identified with all elements of $\{\psi_2, \rho_r, \sigma_r^2\}$; one of these three parameters must be fixed in order to estimate the others (e.g. [Iskrev, 2010](#), [Komunjer and Ng, 2011](#), [Qu and Tkachenko, 2012, 2017](#), [Kocięcki and Kolaszka, 2023](#)). The researcher must therefore decide whether to fix ψ_2 , ρ_r , or σ_r^2 .

Because these parameters have different units, [Figure 2](#) reports miscalibrations in normalized percent-of-range units. Thus, $x = 0$ means correct calibration, and $x = 0.1$ means that the fixed parameter is set 10% of its plausible range above its baseline value⁴. For each curve, we fix one parameter at nearby values, re-estimate the remaining parameters by MLE, and record the implied value of ψ_1 .

The true value is $\psi_1 = 1.5$. Hence, the vertical distance from 1.5 is the bias induced by miscalibration. Fixing ψ_2 is relatively robust: the curve is nearly flat around zero. Fixing ρ_r or σ_r^2 , in contrast, can generate large changes in ψ_1 even for small normalized perturbations.

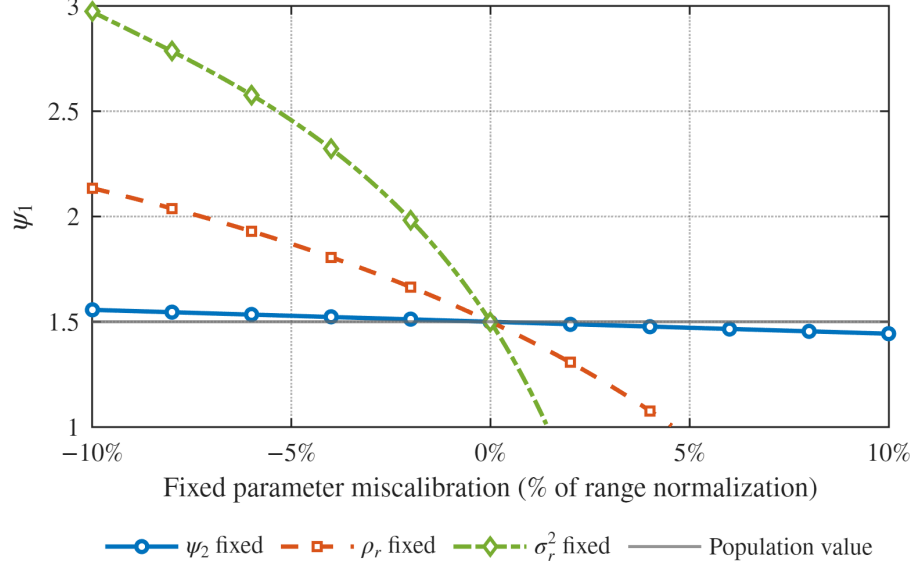
³We use the calibration in [An and Schorfheide \(2007\)](#) as the data-generating process:

$$(\rho, \tau, \kappa, \psi_1, \psi_2, \rho_r, \rho_g, \rho_z, \sigma_r^2, \sigma_g^2, \sigma_z^2) = (0.9975, 2, 0.33, 1.5, 0.125, 0.75, 0.95, 0.9, 0.04, 0.36, 0.09).$$

⁴To compute the normalization ranges we use the priors for ψ_2 , ρ_r , and σ_r^2 in [An and Schorfheide \(2007\)](#). We compute its respective 0.5% and 99.5% quantiles. Obtaining:

$$\psi_2 \in [0, 0.98], \quad \rho_r \in [0, 1], \quad \sigma_r^2 \in [0, 0.35].$$

FIGURE 2. Value of ψ_1 as a function of miscalibration in the fixed parameter



These two examples motivate the criterion developed below. Full re-estimation curves are informative, but they are costly to compute, difficult to compare when the target or the calibrated block is multidimensional, and sensitive to the units of the fixed parameters. The statistic K_S replaces these curves with a local, scalar measure of worst-case sensitivity of the object of interest to calibration error.

3. FRAMEWORK AND PARTITION SELECTION METHODOLOGY

Let $\eta \in \mathcal{N} \subset \mathbb{R}^{d_\eta}$ denote the structural parameter vector and let $\gamma = \Gamma(\eta) \in \mathbb{R}^{d_\gamma}$ denote the object of interest. A partition is an index set $S \subseteq \{1, \dots, d_\eta\}$ selecting which components of η are estimated. Define

$$\alpha := (\eta_j)_{j \in S}, \quad \beta := (\eta_j)_{j \in S^c},$$

so that $\eta = (\alpha, \beta)$, where α denotes the estimated parameters and β the calibrated (fixed) parameters⁵. For a given partition, fixing β and solving the minimum-distance problem yields $\alpha(\beta; S)$, which in turn induces the partition-specific map

$$\gamma(\beta; S) = \Gamma(\alpha(\beta; S), \beta).$$

This induced map is the central object for partition selection: it describes how the object of interest changes when the fixed parameters are perturbed and the remaining parameters are re-estimated.

Our comparison is local. Partitions are evaluated around a reference point η^* , typically obtained from a first-stage estimation, which provides the common baseline at which sensitivities and admissibility are assessed.

The goal of this section is to compare these induced maps across partitions and select the one that is least-sensitive to local miscalibration of the fixed parameters. Formally, after defining the admissible set of partitions $\mathcal{S}_{\text{AD}}$, and the sensitivity statistic K_S , we choose

$$S^* \in \arg \min_{S \in \mathcal{S}_{\text{AD}}} K_S.$$

The rest of the section builds the ingredients of this criterion step by step. Subsection 3.1 introduces the conditional minimum-distance problem, the induced map $\gamma(\beta; S)$, and the reference point η^* . Subsection 3.2 derives the sensitivity statistic K_S . Subsection 3.3 defines the admissible set, $\mathcal{S}_{\text{AD}}$, and the least-sensitive partition S^* .

3.1 Framework

We study the case in which a structural economic model implies a system of d_g equations

$$g(\eta) : \mathcal{N} \longrightarrow \mathcal{G} \subseteq \mathbb{R}^{d_g}.$$

⁵Equivalently, S induces $\mathcal{N} = \mathcal{A} \times \mathcal{B}$ with $\mathcal{A} := \prod_{j \in S} \mathcal{N}_j$ and $\mathcal{B} := \prod_{j \in S^c} \mathcal{N}_j$, where $\mathcal{N}_j$ is the parameter space associated with the j th element of η .

The identified set is defined by all structural parameters that solve the model equations:

$$\mathcal{N}_0 := \{\eta \in \mathcal{N} \mid g(\eta) = 0\}. \quad (1)$$

Throughout the paper we assume the following conditions about the system of equations and the object of interest are satisfied.

ASSUMPTION 1. *i)* There exists a true population parameter $\eta_0 \in \mathcal{N}_0$; *ii)* $g(\alpha, \beta; S)$ is twice continuously differentiable in (α, β) ; *iii)* $\Gamma(\alpha, \beta)$ is continuously differentiable.

Given β , the conditional minimum-distance solution for the estimated parameters α is given by

$$\alpha(\beta; S) \in \arg \min_{\alpha \in \mathcal{A}} g(\alpha, \beta)' W g(\alpha, \beta), \quad (2)$$

where W is a positive-definite weighting matrix.

To compute the sensitivity statistics and the set of admissible partitions, we work around a *reference point* η^* , which may differ from the true parameter η_0 . Typically, it is obtained from a first-stage estimation. For any partition we write $\eta^* = (\alpha^*, \beta^*)$ for its corresponding relabeled components. See Section 5.1 for a detailed discussion on the reference point.

ASSUMPTION 2. *(i)* The reference point η^* is an interior point of $\mathcal{N}$; *(ii)* the reference point satisfies $\eta^* \in \mathcal{N}_0$.

Throughout the main text, we focus on cases in which the system of equations $g(\cdot)$ does not depend on the partition S , except through the relabeling of coordinates induced by the split (α, β) . This condition is satisfied in many applications of GMM, CMD, SMM, and II. There are, however, important cases in which the system of equations does depend on the partition. A leading example is MLE: deciding which parameters are estimated and which are fixed changes the score equations used in the estimation problem. In Appendix B.1 we extend the methodology to the MLE case.

3.2 Normalization and Sensitivity Statistic

We now formalize the local sensitivity measure used to compare candidate partitions. We proceed in three steps: derive how the MD solution responds to perturbations in the fixed parameters, translate this into sensitivity of the object of interest, and normalize the result so it can be compared across partitions.

The Jacobians of the system of equations, evaluated at the reference point, are denoted by:

$$G_\alpha := \frac{\partial g(\alpha^*, \beta^*)}{\partial \alpha}, \quad G_\beta := \frac{\partial g(\alpha^*, \beta^*)}{\partial \beta}.$$

Throughout the paper, we suppress the dependence of these Jacobians on the reference point and on the partition S .

LEMMA 1. *Suppose that Assumptions 1 and 2 hold, and the Jacobian for the estimated parameters satisfies $\text{Rank}(G_\alpha) = |S|$. Then,*

$$\frac{\partial \alpha(\beta^*; S)}{\partial \beta} = - (G'_\alpha W G_\alpha)^{-1} G'_\alpha W G_\beta. \quad (3)$$

Accordingly, by the chain rule,

$$\frac{\partial \gamma(\beta^*; S)}{\partial \beta} = \frac{\partial \Gamma}{\partial \alpha} \frac{\partial \alpha(\beta^*; S)}{\partial \beta} + \frac{\partial \Gamma}{\partial \beta}, \quad (4)$$

where $\partial \Gamma / \partial \alpha$ and $\partial \Gamma / \partial \beta$ are evaluated at (α^*, β^*) .

PROOF. This follows directly from applying the Implicit Function Theorem (IFT) to the first-order conditions of the MD problem in (2). $\square$

Jørgensen (2023) derives equivalent expressions in the GMM setting. Since the derivative of the object of interest depends on the units of the fixed parameters, we normalize them using the diagonal matrix

$$\Sigma_S := \text{diag}((\Delta_j)_{j \in S^c}),$$

where Δ_j determines the scale used for parameter j . Section 5.4 provides further discussion on how to construct it in practice.

DEFINITION 1. We define the sensitivity statistic as

$$K_S(\eta^*, \Sigma_S) := \sqrt{|S^c|} \cdot \left\| \frac{\partial \gamma(\beta^*; S)}{\partial \beta} \Sigma_S \right\|_2. \quad (5)$$

Here, $\|A\|_2 = \sigma_{\max}(A)$ denotes the spectral norm, where $\sigma_{\max}(A)$ is the largest singular value of A . In Section 4, we provide the formal justification for using K_S : we show that K_S is proportional to the worst-case bias induced by a local miscalibration. Equivalently, K_S can be read as the largest first-order change in the object of interest generated by a normalized local perturbation of the fixed parameters, so it summarizes worst-case normalized bias in a single scalar.

Note that K_S depends on the reference point η^* , because the derivative $\partial \gamma(\beta^*; S)/\partial \beta$ is obtained from the IFT using Jacobians evaluated at η^* . It also depends on the normalization matrix.

Under range normalization, Σ_S plays two roles: (i) it removes dependence on arbitrary units of measurement, and (ii) it incorporates external information about plausible parameter values through bounds, priors, or confidence intervals. As a result, K_S captures both the model-implied sensitivity of γ to β and the uncertainty surrounding β^* .

EXAMPLE. Suppose both the object of interest and the fixed parameter are scalars, and let $S^c = \{j\}$. Under range normalization, the sensitivity statistic reduces to

$$K_S = \underbrace{\left| \frac{\partial \gamma(\beta^*; S)}{\partial \beta} \right|}_{\text{Model-implied sensitivity}} \cdot \underbrace{(\eta_{j,\max} - \eta_{j,\min})}_{\text{Uncertainty about fixed parameter}}. \quad (6)$$

Thus, the statistic is simply the product of two components: (i) the sensitivity of the object of interest to the parameter, and (ii) the uncertainty surrounding that parameter's value.

3.3 Admissible Partitions and Least-sensitive Partition

This subsection completes the partition-selection rule. We first define the admissible class of partitions and then select, within that class, the partition that minimizes the sensitivity statistic K_S .

Admissibility imposes three requirements: the object of interest must depend nontrivially on the estimated parameters, the estimated parameters must be conditionally locally identified, and the estimation error in the object of interest must vanish in mean square as n grows.

ASSUMPTION 3 (Admissibility). The partition, S , and the reference point, η^* , satisfy:

- i) Non-trivial partition: $S \in \mathcal{S}$ where $\mathcal{S}$ is

$$\mathcal{S} := \left\{ S \subseteq \{1, \dots, d_\eta\} : 0 < |S|, \frac{\partial \Gamma(\alpha^*, \beta^*)}{\partial \alpha} \neq 0 \right\}.$$

- ii) Conditional local identification: $\text{rank}(G_\alpha) = |S|$.

- iii) Declining second moments of the estimation error:

$$\mathbb{E} \left[\|\hat{\gamma}(\beta^*; S) - \gamma(\beta^*; S)\|_2^2 \right] = o(1). \quad (7)$$

Assumption 3.i) ensures the object of interest is not only (locally) explained by the fixed parameters. In the case where a particular structural parameter is the object of interest it would rule out partitions where that parameter is fixed.

Assumption 3.ii) would be the typical regular local identification rank condition for the estimated parameters.

Third, Section 4 shows that the least-sensitive partition minimizes worst-case local bias and, under an additional higher-order condition, also minimizes worst-case mean squared error asymptotically. For this MSE interpretation, the sampling variation in the estimated object of interest must be asymptotically negligible, so that the criterion is driven by calibration bias. We therefore restrict attention to partitions satisfying Assumption 3.iii).

Finally, let $\mathcal{R}$ encode any hard researcher restrictions (for example, parameters that must always be estimated or always fixed, or upper and lower bounds on the

number of estimated parameters). We define the admissible set as

$$\mathcal{S}_{\text{AD}}(\eta^*) := \{S \in \mathcal{R} \text{ s.t. Assumption 3 holds}\}.$$

Given a reference point η^* , and a normalization matrix Σ_S we evaluate the sensitivity statistic $K_S(\eta^*, \Sigma_S)$ for each partition $S \in \mathcal{S}_{\text{AD}}(\eta^*)$ and choose a partition with the smallest value.

DEFINITION 2 (Least-sensitive partition). Conditional on the reference point η^* , a least-sensitive partition is any solution to

$$S^*(\eta^*, \{\Sigma_S\}) \in \arg \min_{S \in \mathcal{S}_{\text{AD}}(\eta^*)} K_S(\eta^*, \Sigma_S).$$

4. JUSTIFICATION OF THE PARTITION SELECTION METHOD

This section first gives a local minimax-bias justification and then derives an asymptotic worst-case-MSE interpretation. For a given admissible partition S , we interpret the choice of partition through a minimax criterion over a local neighborhood of miscalibrations, and relate this perspective to [Bonhomme and Weidner \(2022\)](#)'s approach to local robust estimation under misspecification.

For a fixed partition S , define the local neighborhood around β^* by

$$B_\epsilon(\beta^*; S) := \left\{ \beta_0 \in \mathcal{B} \text{ s.t. } \|\Sigma_S^{-1}(\beta^* - \beta_0)\|_2 \leq \epsilon \sqrt{|S^c|} \right\}. \quad (8)$$

The factor $\sqrt{|S^c|}$ scales the radius by the number of fixed parameters, so that local perturbations are comparable across partitions with different cardinalities. Thus, ϵ denotes the magnitude of the normalized miscalibration of a single fixed parameter. For a given ϵ -neighborhood, define the worst-case bias by

$$\text{WorstBias}(\beta^*, \epsilon; S) := \max_{\beta_0 \in B_\epsilon(\beta^*; S)} \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2. \quad (9)$$

The following lemma connects worst-case bias to the sensitivity statistic K_S .

LEMMA 2. *Suppose Assumptions 1 and 2 hold, and $S \in \mathcal{S}_{\text{AD}}(\eta^*)$. The worst-case bias and the sensitivity statistic are locally proportional:*

$$\text{WorstBias}(\beta^*, \epsilon; S) = \epsilon K_S + o(\epsilon). \quad (10)$$

PROOF. See Appendix C. □

ASSUMPTION 4. The optimization problem $\min_{S \in \mathcal{S}_{AD}} K_S$ has a unique solution S^* .

The next theorem shows that minimizing K_S is equivalent to minimizing worst-case local bias.

THEOREM 3. Suppose Assumptions 1, 2, and 4 hold. Then there exists a scalar $\mathcal{E} > 0$ such that for all $0 < \epsilon \leq \mathcal{E}$,

$$S^* := \arg \min_{S \in \mathcal{S}_{AD}} K_S = \arg \min_{S \in \mathcal{S}_{AD}} \text{WorstBias}(\beta^*, \epsilon; S). \quad (11)$$

PROOF. See Appendix C. □

In the boundary case where all parameters are estimated ($|S^c| = 0$), no parameters are subject to calibration error. The worst-case bias is exactly zero, making the full-estimation partition strictly preferred under our criterion (for full estimation $K_S = 0$) whenever it is empirically admissible.

Theorem 3 also yields a worst-case mean squared error interpretation. Let

$$\begin{aligned} MSE(\beta^*, \beta_0, S) &:= \mathbb{E} [(\hat{\gamma}(\beta^*) - \gamma(\beta_0))' (\hat{\gamma}(\beta^*) - \gamma(\beta_0))] \\ &= \text{tr} \{ \text{Var}(\hat{\gamma}(\beta^*)) \} + \|\gamma(\beta^*) - \gamma(\beta_0)\|_2^2 \\ &\quad + 2(\gamma(\beta^*) - \gamma(\beta_0))' (\mathbb{E}[\hat{\gamma}(\beta^*)] - \gamma(\beta^*)) \\ &\quad + \|\mathbb{E}[\hat{\gamma}(\beta^*)] - \gamma(\beta^*)\|_2^2, \end{aligned} \quad (12)$$

where $\mathbb{E}[\cdot]$ is taken under the fixed sampling law governing $\hat{\gamma}(\beta^*; S)$.

When $\hat{\gamma}(\beta^*; S)$ satisfies the decay condition in (7), namely

$$\mathbb{E} [\|\hat{\gamma}(\beta^*; S) - \gamma(\beta^*; S)\|_2^2] = o(1),$$

the sampling component of the target estimator is asymptotically negligible. Hence, for fixed ϵ ,

$$MSE(\beta^*, \beta_0, S) = \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2^2 + o(1), \quad (13)$$

uniformly over $\beta_0 \in B_\epsilon(\beta^*; S)$.

This decay requirement is closely related to weak identification. Section 5.3 explains how we operationalize this restriction using spectral Jacobian information, following [Antoine and Renault \(2024\)](#) and [Forneron \(2024\)](#).

THEOREM 4. *Under Assumptions 1, 2, and 4, the least-sensitive partition minimizes, asymptotically, the mean squared error under worst-case local miscalibration. Specifically, there exists a scalar $\mathcal{E} > 0$ such that for all $0 < \epsilon \leq \mathcal{E}$,*

$$\min_{S \in \mathcal{S}_{AD}(\eta^*)} \max_{\beta_0 \in B_\epsilon(\beta^*; S)} \text{MSE}(\beta^*, \beta_0, S) = \text{WorstBias}(\beta^*, \epsilon; S^*)^2 + o(1). \quad (14)$$

PROOF. See Appendix C. □

This interpretation is closely related to [Bonhomme and Weidner \(2022\)](#)’s approach to robust estimation under misspecification.⁶ In both frameworks, the objective can be viewed as minimizing a worst-case mean squared error. The main difference is the decision variable: [Bonhomme and Weidner \(2022\)](#) optimize over influence functions, whereas we optimize over parameter partitions. A second difference is the asymptotic regime. Their misspecification neighborhoods shrink with the sample size, so bias and variance are asymptotically comparable, whereas in our setting the neighborhood is not indexed by n , so the asymptotic MSE is driven by the bias component.

A natural alternative to our sensitivity criterion is to select the partition by minimizing an MSE criterion. Nevertheless, implementing an MSE ranking would require a credible estimate of the full joint covariance matrix of all moments or reduced-form estimates. This is often unavailable when the moments are constructed from multiple datasets, as in many CMD applications. One could use an MSE-based criterion when this covariance matrix is available, or when a reliable joint bootstrap is feasible. In the [Nakamura and Steinsson \(2018\)](#) application, however, different datasets are used to compute the reduced-form parameters entering the CMD estimation, which motivates our sensitivity-based approach.

⁶Their notion of misspecification is more general than the classical formulation considered here and includes our setting as a special case.

REMARK 1 (**Why a local analysis?**). *First*, local methods are typically much easier to implement than global sensitivity methods, which is important in structural applications where each re-estimation can be costly. *Second*, anchoring the analysis at a reference point is natural empirically, since applied work almost always begins with a first-stage estimation that delivers a candidate value for η^* . *Third*, many structural analyses proceed under an implicit assumption of *local* miscalibration: the calibrated (fixed) parameters are believed not to be too far from their population values, since otherwise counterfactual and policy conclusions would be difficult to interpret. This perspective is plausible in some applications—for example, for parameters such as discount factors that are often thought to lie in a relatively narrow range—but it need not hold universally. If η^* is far from η_0 , a local analysis becomes uninformative, and global sensitivity tools are more appropriate.

REMARK 2 (**Using the selected partition under recalibration**). After selecting the least-sensitive partition $S^*(\eta^*)$, the researcher will often want to use it together with an alternative calibration $\tilde{\beta} \neq \beta^*$. Keeping the selected partition fixed, this leads to the updated point

$$\tilde{\eta} := (\alpha(\tilde{\beta}; S^*(\eta^*)), \tilde{\beta}).$$

A natural question is whether the same partition remains optimal when the criterion is re-evaluated at $\tilde{\eta}$. Appendix A.3 studies this issue and gives sufficient conditions under which a nearby recalibration does not change the selected partition. In particular, it shows that if $\tilde{\eta}$ remains sufficiently close to η^* , then

$$S^*(\tilde{\eta}) = S^*(\eta^*).$$

This provides a formal justification for selecting the partition at a reference point and then using it with nearby alternative calibrations.

5. ESTIMATION AND IMPLEMENTATION

This section describes how to implement the partition-selection procedure in finite samples. We begin by presenting the full algorithm.

Algorithm 1 Least-Sensitive Partition

1: **Input:** $(S_1, \beta_1), \mathcal{R}, \{\Sigma_S\}_{S \in \mathcal{S}}, \widehat{g}, \Gamma, \widehat{W}, n$

2: **Step 1: First-stage estimation**

3: Compute

$$\widehat{\alpha}(\beta_1; S_1) \in \arg \min_{\alpha} \widehat{g}(\alpha, \beta_1; S_1)' \widehat{W} \widehat{g}(\alpha, \beta_1; S_1)$$

4: Set $\widehat{\eta} := (\widehat{\alpha}(\beta_1; S_1), \beta_1)$

5: **Step 2: Derivatives at reference point**

6: Compute

$$\widehat{G}_{\eta} = \left. \frac{\partial \widehat{g}(\eta)}{\partial \eta'} \right|_{\widehat{\eta}}, \quad \widehat{\Gamma}_{\eta} = \left. \frac{\partial \Gamma(\eta)}{\partial \eta'} \right|_{\widehat{\eta}}$$

7: **for** each $S \in \mathcal{S} \cap \mathcal{R}$ **do**

8: Compute singular values of $\widehat{W}^{1/2} \widehat{G}_{\alpha, S}$

9: Define $\widehat{r}_S = \#\{j : \widehat{\sigma}_{j, S} > \tau_n\}$

10: **if** $\widehat{r}_S = |S|$ **then**

11: Compute

$$\widehat{D}_{\alpha\beta, S} = -(\widehat{G}'_{\alpha} \widehat{W} \widehat{G}_{\alpha})^{-1} \widehat{G}'_{\alpha} \widehat{W} \widehat{G}_{\beta}$$

12: Compute

$$\widehat{D}_{\gamma\beta, S} = \widehat{\Gamma}_{\alpha} \widehat{D}_{\alpha\beta, S} + \widehat{\Gamma}_{\beta}$$

13: Compute sensitivity

$$\widehat{K}_S = \sqrt{|S^c|} \left\| \widehat{D}_{\gamma\beta, S} \Sigma_S \right\|_2$$

14: **end if**

15: **end for**

16: **Step 5: Selection**

17:

$$\widehat{S}^* \in \arg \min_S \widehat{K}_S$$

18: **Output:** $\widehat{S}_{AD}, \{\widehat{K}_S\}, \widehat{S}^*$

The rest of this section discusses the three implementation choices that enter the algorithm: how to obtain the reference point, how to construct the admissible set of partitions, and how to choose the normalization matrix used in the sensitivity computation.

Formal results on the consistency of the estimated least-sensitive partition are given in Appendix A.5.

5.1 First-stage estimation and the reference point

Starting from a first-stage calibration (S_1, β_1) , the reference point is

$$\hat{\eta} := (\hat{\alpha}(\beta_1; S_1), \beta_1),$$

where $\hat{\alpha}(\beta_1; S_1)$ is the minimum-distance estimate obtained under the first-stage partition and calibration. We assume that the sample reference point satisfies

$$\hat{\eta} \xrightarrow{p} \eta^*.$$

A primitive condition ensuring that $\eta^* \in \mathcal{N}_0$, as required in Assumption 2, is that the first-stage calibration is compatible with the population equations. That is, there exists $\alpha \in \mathcal{A}$ such that

$$g(\alpha, \beta_1; S_1) = 0. \tag{15}$$

This assumption that the reference point lies in the identified set is crucial for three reasons. First, it guarantees *comparability* across partitions: for any candidate partition S calibrating the same β^* , the conditional MD solution $\alpha(\beta^*; S)$ coincides with α^* . This ensures that the points at which the local derivatives are evaluated for different MD problems are comparable. Second, it allows us to *avoid re-estimating* the model for every candidate partition. Third, it allows us to omit the Hessian terms in the IFT formula, which are asymptotically negligible at a reference point in the identified set. The asymptotic justification of the direct derivative computation, which is detailed in the next subsection, relies on these properties.

In overidentified GMM or CMD settings, the condition in (15) is empirically testable. One can use identification-robust calibration-validity tests, as in [Alegre and Escanciano \(2023\)](#), or related weak-identification robust methods such as [Forneron \(2024\)](#).

5.2 Derivative computation without re-estimation

The statistic $\hat{D}_{\alpha\beta,S}$, proposed in the algorithm, should be interpreted as a direct plug-in computation of the local derivative, not as the exact finite-sample derivative of every partition-specific re-estimation map. To see the distinction, fix a candidate partition S and write the reference point under the S -specific ordering as

$$\hat{\eta} = (\hat{\alpha}, \hat{\beta}).$$

A literal finite-sample computation of the slope of the re-estimation map would first solve

$$\hat{\alpha}(\hat{\beta}; S) \in \arg \min_{\alpha \in A_S} \hat{g}(\alpha, \hat{\beta}; S)' \widehat{W} \hat{g}(\alpha, \hat{\beta}; S),$$

and then differentiate the corresponding first-order conditions with respect to the fixed parameters. Doing this separately for each partition would require re-estimating the model many times. Importantly, $\hat{\alpha}$, the S -coordinates of the reference point, need not coincide with $\hat{\alpha}(\hat{\beta}; S)$, the MD solution obtained after fixing $\hat{\beta}$.

The exact finite-sample derivative would also include additional Hessian terms weighted by the residuals $\hat{g}(\hat{\alpha}(\hat{\beta}), \hat{\beta})$; under Assumption 2, which places the reference point in the identified set, these terms vanish asymptotically.

The algorithm avoids this re-estimation step. Instead, for each partition S , it uses the derivative bundle evaluated once at the common reference point $\hat{\eta}$ and computes

$$\hat{D}_{\alpha\beta}(S) = - \left(\hat{G}_{\alpha}(\hat{\eta}; S)' \widehat{W} \hat{G}_{\alpha}(\hat{\eta}; S) \right)^{-1} \hat{G}_{\alpha}(\hat{\eta}; S)' \widehat{W} \hat{G}_{\beta}(\hat{\eta}; S).$$

The justification is asymptotic. Under the regularity conditions in Appendix A.6, and using the reference point in the identified set, $\hat{D}_{\alpha\beta}(S)$ provides a consistent

asymptotic approximation to the population slope,

$$\left. \frac{\partial \hat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\hat{\beta}} = \hat{D}_{\alpha\beta}(S) + o_p(1), \quad \hat{D}_{\alpha\beta}(S) = \frac{\partial \alpha(\beta^*; S)}{\partial \beta'} + o_p(1).$$

Therefore, re-estimating $\hat{\alpha}(\hat{\beta}; S)$ for every candidate partition would only change the derivative by an $o_p(1)$ term. The same argument applies to the derivative of the target object.

5.3 Practical guidance on the implementation of the admissible set of partitions

The goal of this subsection is to explain how we implement the conditional local identification and declining-variance conditions used in the definition of the admissible set of partitions. We implement both restrictions using singular values of the weighted Jacobian of the estimating equations. The basic idea is that small singular values of $W^{1/2}G_\alpha$ make the inverse problem unstable, which can prevent the variance of the induced estimator of γ from declining with the sample size.

For a candidate partition S , let

$$C_S := (W^{1/2}G_\alpha)^+,$$

where $(\cdot)^+$ denotes the Moore–Penrose generalized inverse. The norm $\|\cdot\|_2$ denotes the spectral norm, i.e., the largest singular value of a matrix. When $W^{1/2}G_\alpha$ has full column rank,

$$\|C_S\|_2 = \frac{1}{\sigma_{\min}\{W^{1/2}G_\alpha\}}. \quad (16)$$

Thus, small singular values of the weighted Jacobian correspond to a large Moore–Penrose inverse and indicate that the partition is close to a weak-identification region.

To keep notation light, we suppress the local-sequence index in the main text.⁷

⁷Strictly speaking, the population Jacobian and the matrix C_S should be indexed by n along the local sequence. Appendix A.4 gives the formal local-sequence argument.

Under local curvature restrictions on g and regularity conditions on Γ , Appendix A.4 shows that if

$$\|C_S\|_2 = o(\sqrt{n}),$$

then

$$\mathbb{E}[\|\hat{\gamma}(\beta^*) - \gamma(\beta^*)\|^2] = o(1), \quad \text{tr}\{\text{Var}(\hat{\gamma}(\beta^*))\} = o(1). \quad (17)$$

Equations (16) and (17) motivate a hard-thresholding rule for the singular values of the estimated weighted Jacobian. For each partition S , let

$$\hat{\sigma}_{1,S} \geq \hat{\sigma}_{2,S} \geq \dots$$

denote the singular values of $\widehat{W}^{1/2}\widehat{G}_{\alpha,S}$. We estimate the conditional rank of partition S by

$$\hat{r}_S = \#\{j : \hat{\sigma}_{j,S} > \tau_n\}, \quad \tau_n = \sqrt{\frac{\log n}{n}}, \quad (18)$$

where the threshold follows [Forneron \(2024\)](#). A partition is kept only if

$$\hat{r}_S = |S|.$$

This rule removes partitions with exact rank failure and partitions whose weighted Jacobian is numerically close to rank deficient. The estimated admissible set is therefore

$$\widehat{\mathcal{S}}_{AD}(\hat{\eta}) = \{S \in \mathcal{S} \cap \mathcal{R} : \hat{r}_S = |S|\}.$$

5.4 Practical Guidance on the Normalization Matrix

The normalization matrix Σ_S defines the metric used to measure local miscalibrations of the fixed parameters. Its purpose is to make perturbations comparable across structural parameters that may be expressed in different units. This subsection explains how we construct this metric, and how we conduct robustness analysis.

Before evaluating candidate partitions, we assign each structural parameter j an interval

$$[\eta_{j,\min}, \eta_{j,\max}].$$

We use the range $\eta_{j,\max} - \eta_{j,\min}$. To obtain the ranges, we can use one of the following options.

REMARK 3 (Sources of Bounds). The intervals $(\eta_{j,\min}, \eta_{j,\max})$ should be chosen before evaluating the candidate partitions and should be based on conservative restrictions. In the applications below we use three sources.

- i) **Parameter-space bounds.** When the model imposes natural bounds on a parameter, we can use those bounds directly.
- ii) **Model-implied economic bounds.** When a parameter is more naturally disciplined through an economic object—for example, the steady-state interest rate—, we impose bounds on that object and map them into the structural parameter, say the discount factor. For example, bounds on steady-state markups can be translated into bounds on the elasticity of substitution across varieties.
- iii) **Well-established external intervals.** When direct external evidence is available, we can use published intervals or widely used conservative quantiles from the relevant literature. For example, in the empirical application we use external evidence on the slope of the Phillips curve given by [Hazell et al. \(2022\)](#).

Because the normalization matrix encodes what the researcher regards as a plausible calibration error, robustness to this choice is an important part of the implementation. The baseline normalization should therefore be interpreted as the main specification, not as the only admissible metric. When several normalizations are equally defensible, the researcher should verify whether the selected partition and the substantive conclusions are stable across them.

A first robustness exercise is to replace the baseline normalization by a family of plausible normalization matrices. Let $\mathcal{M}_S$ denote the set of normalization matrices considered plausible for partition S . Instead of selecting the partition that

minimizes $K_S(\eta^*, \Sigma_S)$ for a single matrix, one can compute

$$S^{rob} \in \arg \min_{S \in \mathcal{S}_{AD}(\eta^*)} \max_{\Sigma_S \in \mathcal{M}_S} K_S(\eta^*, \Sigma_S).$$

A second exercise is diagnostic. For each partition, define the normalized Jacobian

$$A_S := D_{\gamma\beta, S} \Sigma_S, \quad D_{\gamma\beta, S} := \frac{\partial \gamma(\beta^*; S)}{\partial \beta'}.$$

Let v_S be a right singular vector associated with the largest singular value of A_S , equivalently an eigenvector of $A'_S A_S$ associated with its largest eigenvalue. The squared entries of v_S quantify the proportion of the worst-case normalized perturbation allocated to each calibrated parameter.

These contribution measures identify which calibrated parameters drive the sensitivity statistic. A targeted robustness check is then to widen the normalization bounds only for the parameters with the largest contributions and recompute K_S , the selected partition, and the implied worst-case bounds. This stress test asks whether the empirical conclusion depends on tight bounds for the parameters that matter most for worst-case miscalibration. If the selected partition and the substantive conclusions remain stable after widening those influential bounds, the robustness of the result is more credible.

6. EMPIRICAL APPLICATION

[Nakamura and Steinsson \(2018\)](#) provide evidence of monetary non-neutrality using high-frequency responses of real interest rates, expected inflation, and output growth to Federal Reserve announcements. A central implication of their analysis is that monetary policy affects the economy not only through conventional channels, but also by shifting market beliefs about broader economic fundamentals via *information effects*. They study this mechanism by extending a New Keynesian model to incorporate an information effect and estimating the model by classical minimum distance (CMD), matching model-implied impulse response functions (IRFs) to reduced-form IRFs obtained via local projections.

We use their environment as a laboratory for our partition-selection criterion. Our object of interest is the information-effect parameter γ . Throughout, we hold fixed the model, the CMD objective, and the reduced-form moments; what varies across implementations is only the partition of structural parameters into estimated and calibrated (fixed) components. In this setting, the mapping $g(\cdot; S)$ depends on S only through the relabeling of estimated and fixed parameters, while its functional form remains constant across partitions. The number of reduced-form parameters matched in CMD is $d_m = 33$, and the number of structural parameters is $d_\eta = 10$.

This application illustrates why we use the sensitivity-based statistic rather than a plug-in MSE criterion. The CMD reduced-form objects are constructed from different datasets, so the full joint covariance matrix of the stacked reduced-form parameters is not identified. In particular, the cross-covariances across moment blocks estimated from different datasets are not generally identified from the reported reduced-form estimates.

Table 2 reports the structural parameters and the values reported in [Nakamura and Steinsson \(2018\)](#). We evaluate all local objects at the empirical reference point implied by their estimates, and we denote the corresponding estimate of the information effect by $\hat{\gamma}$. [Alegre and Escanciano \(2023\)](#) test the Nakamura–Steinsson benchmark calibration on a feasible subset of moments and do not reject it, supporting its use as the local reference point.

6.1 Informative Range Normalization

In this New Keynesian model, credible external information about the calibrated parameters is available, so we can use it to enrich the sensitivity analysis by tightening the bounds used in the normalization.

Because ad hoc modifications of the normalization ranges could potentially change the analysis in arbitrary ways, it is important to be transparent about how these tighter bounds are chosen. It is also important to conduct robustness checks for any parameter whose normalization plays a central role in determining the selected partition.

TABLE 2. **Estimated and Fixed Parameters in Nakamura and Steinsson (2018)**

<i>Notation</i>	<i>Definition</i>	<i>Value</i>
<i>Fixed parameters</i>		
ρ	<i>Subjective discount factor</i>	0.99
δ	<i>Nominal rigidity</i>	0.75
ω	<i>Elasticity of marginal cost to output</i>	2
θ	<i>Elasticity of substitution across varieties</i>	10
σ	<i>Intertemporal elasticity of substitution</i>	0.50
b	<i>Consumption habit</i>	0.90
<i>Estimated parameters</i>		
ρ_1	<i>Autoregressive first root of monetary shock</i>	0.90
ρ_2	<i>Autoregressive second root of monetary shock</i>	0.79
$\kappa\zeta \times 10^5$	<i>Phillips curve slope</i>	11.2
γ	<i>Information effect</i>	0.675

Notes: In the original paper, three additional fixed structural parameters (the inflation target shock, the endogenous feedback of the Taylor rule, and the coefficient of lagged inflation) are omitted here to streamline the analysis.

We proceed as follows. First, we explain, parameter by parameter, how the informative bounds are chosen. We then present the main results. Finally, we conduct a robustness exercise for the discount-factor bounds in Subsection 6.3, since this parameter is a key driver of the main results.

6.1.1 Discount factor ρ . We follow Schorfheide (2000) and set $\rho \sim \text{Beta}(\mu = 0.993, \sigma = 0.002)$. The normalization bounds are the 1% and 99% quantiles of this prior. Similar tight priors are standard in estimated DSGE work; see, e.g., Lubik and Schorfheide (2005), Smets and Wouters (2007), and An and Schorfheide (2007). This reflects the view that the discount factor is tightly pinned down by the steady-state relationship with the steady-state interest rate.

6.1.2 *Nominal rigidity* δ . We follow [Smets and Wouters \(2007\)](#) and set $\delta \sim \text{Beta}(\mu = 0.5, \sigma = 0.1)$, taking the 0.5% and 99.5% quantiles as bounds. Closely related priors are used by [Rabanal and Rubio-Ramírez \(2005\)](#), and [Lubik and Schorfheide \(2005\)](#).

6.1.3 *Intertemporal elasticity parameter* σ . Following [Smets and Wouters \(2007\)](#), we place a Normal prior on $1/\sigma$:

$$\frac{1}{\sigma} \sim \mathcal{N}(1.5, 0.375). \quad (19)$$

We take the 2.5% and 97.5% quantiles of $1/\sigma$ and invert them to obtain the implied bounds $[\sigma_{\min}, \sigma_{\max}]$ (noting that inversion typically yields an asymmetric interval in σ).

6.1.4 *Habit formation* b . As in [Smets and Wouters \(2007\)](#), we set $b \sim \text{Beta}(\mu = 0.7, \sigma = 0.1)$ and use the 1% and 99% quantiles as bounds. These values capture the empirically plausible region in which habits provide internal propagation without implying nearly unit root consumption dynamics.

6.1.5 *Monetary shock persistence* ρ_1 and ρ_2 . For the autoregressive roots governing the monetary shock, we use the natural positive bounds implied by stationarity and the model's parametric restrictions.

6.1.6 *Elasticity of marginal cost to output* ω . We follow [Smets and Wouters \(2007\)](#) by anchoring ω to more primitive objects: the labor share a and the Frisch elasticity v . Using the model-implied mapping $\omega = (v^{-1} + 1 - a)/a$, we propagate uncertainty from (a, v) into ω . Concretely, we set $a \in \{1/2, 3/4\}$ and

$$v \sim \mathcal{N}(2, 0.75), \quad (20)$$

and compute the implied bounds on ω from conservative quantiles of v (2.5% and 97.5%) and the endpoints for a . This construction yields economically interpretable variation in ω while remaining grounded in well-studied labor-supply and factor-share objects.

6.1.7 *Slope of the Phillips curve.* We discipline the range for the Phillips-curve slope using the full-sample tradeable-demand IV estimate from [Hazell et al. \(2022\)](#). Since the units differ, we map it into [Nakamura and Steinsson \(2018\)](#) units using the following model-implied relationship:

$$\kappa\zeta = \frac{v}{1 + \omega\theta} \kappa^{Hazell},$$

where v is the Frisch labor-supply elasticity, ω is the elasticity of marginal cost with respect to output, and θ is the elasticity of substitution across varieties. Using $v = 1$, which corresponds to a labor share of $2/3$ when $\omega = 2$, the confidence interval yields

$$\kappa\zeta \in [6.30, 52.73] \times 10^{-5}.$$

6.1.8 *Elasticity of substitution across varieties θ .* We use the steady-state relationship between the markup M and θ ,

$$M = \frac{\theta}{\theta - 1}, \tag{21}$$

and choose $M \in [1.05, 1.30]$, i.e. markups between 5% and 30%. We then map this interval into bounds for θ . This pins down θ using a primitive object (markups) with a clear economic interpretation and avoids implausible values that would imply either near-perfect competition or excessively high steady-state profit shares.

6.2 Main Results

Using the hard-thresholding rank estimator proposed in [Section 5.3](#), we estimate the set of admissible partitions. In the present application, this procedure yields a total of 60 admissible partitions. For readability, we present a representative subset in the main text; full tables are reported in [D](#).

For each admissible partition S , we compute the sensitivity statistic $\hat{K}(\hat{\beta}; S)$, which measures the magnitude of worst-case local bias in γ induced by small miscalibrations. To aid interpretation, we report 5% *worst-case bounds*, defined as $\hat{\gamma} \pm 0.05 \hat{K}(\hat{\beta}; S)$, which correspond to a first-order approximation of the worst-case bias of size $\epsilon = 5\%$.

TABLE 3. **Ranges used for normalization of parameter units**

<i>Parameter</i>	<i>Definition</i>	$\eta_{j,\min}$	$\eta_{j,\max}$
ρ	<i>Subjective discount factor</i>	0.987	0.997
δ	<i>Nominal rigidity</i>	0.25	0.75
ω	<i>Elasticity of marginal cost to output</i>	0.72	4.77
θ	<i>Elasticity of substitution across varieties</i>	4.3	21
σ	<i>Intertemporal elasticity of substitution</i>	0.447	1.307
ρ_1	<i>Autoregressive first root of monetary shock</i>	0	1
ρ_2	<i>Autoregressive second root of monetary shock</i>	0	1
$\kappa\zeta \times 10^5$	<i>Phillips curve slope</i>	6.3	52.73
b	<i>Consumption habit</i>	0.36	0.94

Table 4 reports these statistics for the representative subset of admissible partitions. For comparison, we also report the implementation in Nakamura and Steinsson (2018) (labeled *Original*), even though it is not admissible under our rank hard-thresholding criterion.

Table 4 reports the sensitivity statistic $\widehat{K}(\widehat{\beta}; S)$ for a representative subset of admissible partitions. Across the full set of admissible partitions, ω , θ , and the Phillips-curve slope $\kappa\zeta$ are never estimated. This suggests that these parameters are weakly identified in the local rank sense used to construct the admissible set.

Conditional on this admissibility restriction, the choice of partition matters substantially. The least-sensitive partition, S^* , estimates $(\delta, \sigma, \rho_1, \rho_2, \gamma, b)$ and fixes $(\rho, \omega, \theta, \kappa\zeta)$. It delivers $\widehat{K}(\widehat{\beta}; S^*) = 0.06$, so a 5% normalized miscalibration generates only the local worst-case interval $[0.67, 0.68]$. At the opposite end of the admissible set, the statistic rises to 11.07, producing the much wider interval $[0.12, 1.23]$. The original Nakamura–Steinsson implementation is not admissible

TABLE 4. **Admissible partitions and sensitivity statistic $\hat{K}(\hat{\beta}; S)$**

<i>Name</i>	<i>Estimated</i>	<i>Calibrated</i>	$\hat{K}(\hat{\beta}; S)$	5% bounds
<i>Original</i>	$(\rho_1, \rho_2, \gamma, \kappa\zeta)$	$(\rho, \delta, \omega, \theta, \sigma, b)$	1.82	[0.58, 0.768]
S^*	$(\delta, \sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \kappa\zeta)$	0.06	[0.67, 0.68]
S_2	$(\rho, \delta, \rho_1, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \kappa\zeta)$	0.32	[0.66, 0.69]
S_3	$(\rho, \rho_1, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \sigma, \kappa\zeta)$	0.36	[0.66, 0.70]
S_4	$(\rho, \delta, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \rho_1, \kappa\zeta)$	0.38	[0.66, 0.70]
S_{12}	$(\rho, \delta, \sigma, \rho_1, \rho_2, \gamma)$	$(\omega, \theta, \kappa\zeta, b)$	1.05	[0.63, 0.73]
S_{14}	$(\rho, \sigma, \rho_1, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \kappa\zeta)$	1.19	[0.62, 0.74]
S_{25}	$(\delta, \rho_1, \rho_2, \gamma)$	$(\rho, \omega, \theta, \sigma, \kappa\zeta, b)$	1.82	[0.59, 0.77]
S_{39}	$(\rho, \sigma, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \rho_1, \kappa\zeta)$	3.70	[0.49, 0.86]
S_{46}	$(\rho, \delta, \sigma, \gamma)$	$(\omega, \theta, \rho_1, \rho_2, \kappa\zeta, b)$	9.25	[0.22, 1.14]
S_{60}	(γ)	$(\rho, \delta, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta, b)$	11.07	[0.12, 1.23]

under the rank criterion and has $\hat{K} = 1.82$. Hence the original partition is not in the extreme tail, but it is far from the least-sensitive admissible partition.

This heterogeneity is not well summarized by the number of estimated parameters. Among the largest admissible partitions, changing the identity of a single calibrated parameter moves the statistic from 0.06 to values above one. Fixing ρ , as in S^* , is much less costly than fixing σ , b , or δ . The contribution table clarifies why: in the low- and middle-sensitivity partitions, worst-case bias is typically concentrated in one preference or propagation parameter, such as ρ , σ , δ , or b . Robustness therefore depends less on estimating many parameters per se than on estimating the parameters through which calibration errors are most strongly transmitted to γ .

The role of the discount factor illustrates this point. In S^* , almost all remaining sensitivity comes from the calibrated discount factor ρ . This might suggest that the result is mechanically driven by the tight normalization range imposed on ρ . The comparison across partitions argues against this interpretation. Several low-sensitivity alternatives, such as S_2 , S_3 , and S_4 , estimate ρ and still deliver small values of $\hat{K}$. Moreover, the robustness exercise in Table 6 shows directly how much the discount-factor range can be widened before S^* ceases to be the least-sensitive partition. Thus, the tight bound on ρ helps explain the ranking, but it does not trivially determine the conclusion.

Finally, the high-sensitivity tail has a clear economic structure. Extreme fragility appears when the persistence of the monetary-policy shock is imposed by calibration rather than disciplined by the data. Once both persistence roots are fixed, $\hat{K}$ jumps into the range between roughly 9 and 11, and the contribution decomposition shows that ρ_1 , with a smaller role for ρ_2 , accounts for almost all of the statistic. Thus, for the information-effect parameter, the key positive prescription is to estimate the monetary-shock persistence parameters, especially ρ_1 . By contrast, calibrating the weakly identified Phillips-curve and markup-related block $(\omega, \theta, \kappa_\zeta)$ appears comparatively harmless for local robustness. Table 5 decomposes $\hat{K}(\hat{\beta}; S)$ into the relative contribution of each fixed parameter. The main message is that sensitivity is typically highly concentrated: in most partitions, one fixed parameter accounts for almost all of the statistic. In the least-sensitive partition S^* , essentially all residual sensitivity comes from the discount factor ρ , which accounts for 99.6% of the total. In S_2 , the contribution is entirely due to σ . In S_3 , most of the contribution comes from σ (70.7%), with a secondary role for nominal rigidity δ (22.9%). The same concentration appears among admissible partitions that estimate the maximum number of parameters: in S_{12} , the sensitivity is entirely driven by habit formation b , whereas in S_{14} , most of the sensitivity comes from δ , with smaller contributions from ω and θ .

Another important feature is that the parameters that are always fixed because of weak-identification concerns— ω , θ , and κ_ζ —have close to zero contribution to worst-case bias. In particular, the Phillips curve slope κ_ζ contributes essentially zero, and ω and θ are almost never dominant sources of sensitivity.

TABLE 5. **Contribution of fixed parameters to the sensitivity statistic**

<i>Name</i>	ρ	δ	ω	θ	σ	ρ_1	ρ_2	$\kappa\zeta$	γ	b
<i>Original</i>	0.0%	0.0%	0.0%	0.0%	0.0%	-	-	-	-	100.0%
S^*	99.6%	-	0.4%	0.0%	-	-	-	0.0%	-	-
S_2	-	-	0.0%	0.0%	100.0%	-	-	0.0%	-	-
S_3	-	22.9%	2.0%	4.3%	70.7%	-	-	0.2%	-	-
S_4	-	-	0.0%	0.0%	87.6%	12.4%	-	0.0%	-	-
S_{12}	-	-	0.0%	0.0%	-	-	-	0.0%	-	100.0%
S_{14}	-	78.0%	6.8%	14.7%	-	-	-	0.5%	-	-
S_{25}	0.0%	-	0.0%	0.0%	0.0%	-	-	0.0%	-	100.0%
S_{39}	-	10.1%	0.9%	1.9%	-	87.1%	-	0.1%	-	-
S_{46}	-	-	0.0%	0.0%	-	80.0%	18.1%	0.0%	-	1.9%
S_{60}	0.0%	0.0%	0.0%	0.0%	0.0%	82.9%	15.0%	0.0%	-	2.1%

Thus, although these parameters must be calibrated, local miscalibration of them would not have large effects on the object of interest. By contrast, in the low- and middle-sensitivity regions of the admissible set, most of the sensitivity comes from preference and propagation parameters such as ρ , σ , δ , and b .

The high-sensitivity tail has a different structure. Once monetary-shock persistence is fixed rather than estimated, the contribution of ρ_1 becomes dominant. In S_{39} , for example, ρ_1 accounts for 87.1% of total sensitivity. When both persistence roots are fixed, as in S_{46} , ρ_1 accounts for 80.0% and ρ_2 for 18.1%. The same pattern appears in the most fragile partitions: in S_{58} , ρ_1 and ρ_2 account for 85.2% and 14.2%, respectively; in S_{59} , for 77.4% and 22.6%; and in S_{60} , for 82.9% and 15.0%.

The economic interpretation is therefore clear: the most fragile partition strategies are those that force the persistence of the monetary shock to come from calibration rather than estimation. From a robustness perspective, the most valuable discipline is to estimate the shock-persistence parameters, especially ρ_1 , while calibration errors in the slope of the Phillips curve, κ_ζ , and other parameters such as ω and θ are comparatively harmless.

Because the estimated admissible set depends on the singular-value cutoff used in the rank estimator, Appendix E.1 reports a robustness exercise with respect to this implementation choice. In that exercise, we replace the baseline cutoff $\tau_n = (\log n/n)^{1/2}$ by $\tau_n(a) = (\log n/n)^a$ for $a \in \{0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$, recompute the admissible set, and rank partitions again by the sensitivity statistic. The least-sensitive partition remains S^* for all thresholds considered. For $a \leq 0.80$, both the admissible set and the top-five ranking are unchanged; more permissive cutoffs admit additional partitions but do not overturn the main ranking.

6.3 Robustness to discount-factor normalization bounds

The discount-factor normalization is a natural place to check robustness. ρ maps into the steady-state annual nominal interest rate r according to

$$\rho = \frac{1}{1 + r/400} \quad \Longleftrightarrow \quad r = 400 \left(\frac{1}{\rho} - 1 \right). \quad (22)$$

Thus, small numerical changes in the range for ρ can correspond to large economic changes in steady-state interest rates.

The exercise below asks how far the discount-factor bound must be relaxed before the partition ranking changes. For each value of $\rho_{\min}$ we recompute the sensitivity statistics for the least-sensitive partition, its closest competitors, and the original partition, and report the annual rate implied by $\rho_{\min}$. The baseline selected partition remains least-sensitive until the lower bound is pushed below roughly $\rho_{\min} = 0.945$. At that point, the implied steady-state annual nominal rate is above 23%, which is difficult to interpret as a plausible benchmark for the economies typically targeted by this class of models.

TABLE 6. Sensitivities with different normalization bounds of the discount factor

ρ_{min}	S^*	S_2	S_3	S_4	<i>Original</i>	<i>Implied S.S. rate</i>
0.990	0.04	0.32	0.36	0.38	1.82	4.04%
0.985	0.07	0.32	0.36	0.38	1.82	6.09%
0.980	0.10	0.32	0.36	0.38	1.82	8.16%
0.975	0.13	0.32	0.36	0.38	1.82	10.26%
0.970	0.16	0.32	0.36	0.38	1.82	12.37%
0.965	0.19	0.32	0.36	0.38	1.82	14.51%
0.960	0.22	0.32	0.36	0.38	1.82	16.67%
0.955	0.24	0.32	0.36	0.38	1.82	18.85%
0.950	0.27	0.32	0.36	0.38	1.82	21.05%
0.945	0.30	0.32	0.36	0.38	1.82	23.28%
0.940	0.33	0.32	0.36	0.38	1.82	25.53%

7. MONTE CARLO SIMULATION

This section uses Monte Carlo simulations to study the finite-sample implications of the partition-selection criterion in the [Nakamura and Steinsson \(2018\)](#) application. For details in the implementation of the Monte Carlo simulation see [E](#).

The goal of the simulations is to assess how informative the K_S -based sensitivity ranking is for finite-sample bias, variance, and MSE. We generate repeated samples, of sizes $n = 150$ and $n = 500$, from a data-generating process (DGP) calibrated to the reference parameter vector used in the empirical exercise. The Monte Carlo setting is useful because it allows us to compute finite-sample distributions and relevant statistics—such as the variance and mean squared error (MSE) of $\hat{\gamma}$ —that are difficult to obtain in the empirical application, where the reduced-form parameters entering the CMD estimation are constructed from different datasets and their joint variance–covariance matrix is not readily available.

The exercise proceeds as follows. For each admissible partition S , we first compute the local worst-case direction at the reference point and then scale the perturbation by the miscalibration size $\varepsilon \in \{5\%, 10\%, 15\%\}$. We then re-estimate the

model under partition S using this approximated worst-case miscalibration. Repeating this procedure across Monte Carlo samples delivers, for each partition, the finite-sample distribution of $\hat{\gamma}$ under its *own* worst-case miscalibration. We summarize these distributions by reporting the variance and the corresponding worst-case MSE, and by plotting histograms that make the differences across partitions visually transparent.

Table 7.: Finite-sample performance by partition, sample size, and miscalibration size

Partition	Estimated parameters	Calibrated parameters	$\varepsilon = 5\%$			$\varepsilon = 10\%$			$\varepsilon = 15\%$		
			Bias	Var.	MSE	Bias	Var.	MSE	Bias	Var.	MSE
Panel A. Simulated sample size $n = 150$											
S^*	$(\delta, \sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \kappa\zeta)$	42.5	4.6	4.7	77.3	4.9	5.5	115.4	5.4	6.7
S_2	$(\rho, \delta, \rho_1, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \kappa\zeta)$	196.2	1.9	5.7	-269.4	1.4	8.6	-458.6	1.4	22.4
S_3	$(\rho, \rho_1, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \sigma, \kappa\zeta)$	303.4	1.5	10.7	-267.2	1.7	8.8	-339.8	1.3	12.9
S_4	$(\rho, \delta, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \rho_1, \kappa\zeta)$	438.8	1.8	21.1	-1867.4	0.6	349.3	-3861.6	1.6	1492.8
S_5	$(\sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \delta, \omega, \theta, \kappa\zeta)$	882.4	1.8	79.7	2094.4	10.3	448.9	2427.0	13.8	602.8
S_6	$(\delta, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \sigma, \kappa\zeta)$	-291.5	1.5	10.0	-576.0	1.7	34.9	-879.5	2.0	79.4
Original	$(\rho_1, \rho_2, \kappa\zeta, \gamma)$	$(\rho, \delta, \omega, \theta, \sigma, b)$	951.7	0.8	91.4	1880.6	0.6	354.3	2389.4	0.3	571.3
Panel B. Simulated sample size $n = 500$											
S^*	$(\delta, \sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \kappa\zeta)$	32.6	1.2	1.4	66.0	1.3	1.8	102.5	1.4	2.5
S_2	$(\rho, \delta, \rho_1, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \kappa\zeta)$	198.7	0.6	4.5	-256.6	0.5	7.0	-453.5	0.5	21.0
S_3	$(\rho, \rho_1, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \sigma, \kappa\zeta)$	301.4	0.6	9.6	-259.5	0.6	7.3	-334.7	0.5	11.7
S_4	$(\rho, \delta, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \rho_1, \kappa\zeta)$	445.6	0.6	20.4	-1857.2	0.2	345.1	-3861.7	3.1	1494.3
S_5	$(\sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \delta, \omega, \theta, \kappa\zeta)$	880.0	0.6	78.1	2051.6	17.0	437.9	2430.0	14.8	605.3
S_6	$(\delta, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \sigma, \kappa\zeta)$	-286.3	0.5	8.7	-569.0	0.5	32.9	-870.6	0.6	76.4
Original	$(\rho_1, \rho_2, \kappa\zeta, \gamma)$	$(\rho, \delta, \omega, \theta, \sigma, b)$	945.7	0.3	89.7	1877.9	0.2	352.9	2392.8	0.1	572.7

Table 7 delivers three main lessons. First, the K_S -based ranking is highly informative about finite-sample worst-case bias. The least-sensitive partition, S^* , has the smallest absolute bias and the lowest MSE for every miscalibration size and for both sample sizes. This is not mechanically imposed by the criterion, which is based on a local worst-case bias approximation rather than on simulated finite-sample MSE. The Monte Carlo results therefore support the interpretation of K_S as a useful finite-sample diagnostic of calibration risk.

Second, the results make clear that sampling variance is not the main source of the differences across partitions. Increasing the sample size from $n = 150$ to $n = 500$ reduces the variance of $\hat{\gamma}$ substantially, but it leaves the miscalibration-induced bias largely unchanged. As a result, larger samples do not eliminate the consequences of fixing sensitive parameters at incorrect values. Instead, they make the bias component more visible. For S^* , the MSE remains small even at $\varepsilon = 15\%$, whereas the MSE of more fragile partitions grows rapidly as the miscalibration size increases.

Third, the deterioration in performance is strongly partition-specific. Partitions such as S_2 , S_3 , and S_6 generate moderate biases relative to the most fragile cases, but they still perform worse than S^* across the table. By contrast, partitions such as S_4 , S_5 , and the original Nakamura–Steinsson implementation can generate very large worst-case biases, especially at $\varepsilon = 10\%$ and $\varepsilon = 15\%$. The original partition is therefore not fragile because of sampling variability; it is fragile because the fixed block contains parameters through which calibration errors are strongly transmitted to γ .

Figure 3 visualizes the same pattern. For very small miscalibrations, the distributions of $\hat{\gamma}$ under the admissible partitions are relatively close to one another, so sampling variation can obscure the gains from reducing worst-case bias. As ε increases, the distributions separate sharply. The original partition moves far away from the true value, while the distribution under S^* remains much closer to it. Increasing the sample size tightens the distributions, but it does not remove the displacement generated by fixing miscalibrated parameters. This explains why the advantage of S^* becomes clearer for larger samples and larger miscalibrations: sampling uncertainty falls, while calibration bias remains.

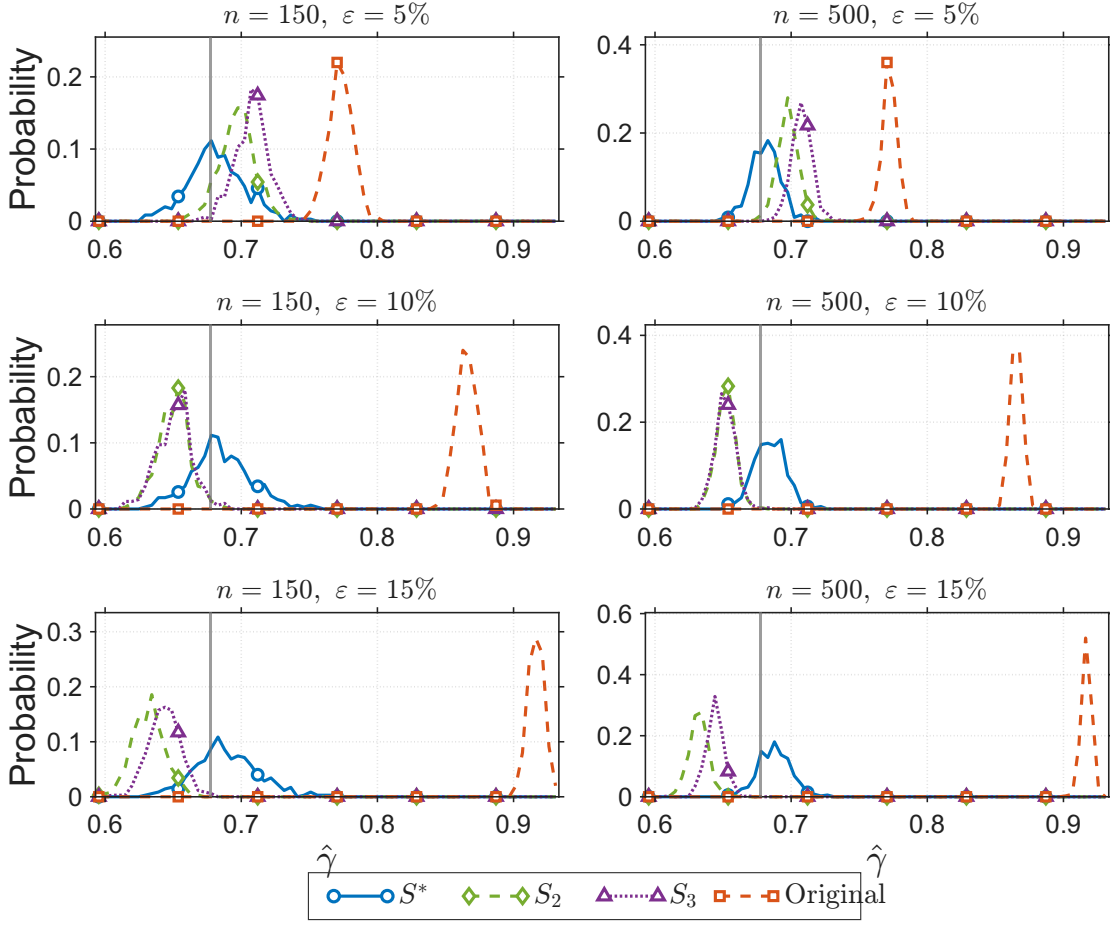


FIGURE 3. Finite-sample distribution of $\hat{\gamma}$ across selected partitions, sample sizes, and miscalibration sizes.

Figure 4 reports the worst-case information effect, γ , as a function of the miscalibration size ϵ for the main partitions. The figure summarizes how the target object responds under increasing local calibration error, highlighting substantial differences in robustness across partitions: the selected partition remains comparatively stable, while the original implementation exhibits a pronounced deterioration as ϵ increases, with intermediate partitions lying between these extremes. Overall, the plot provides a direct visual counterpart to the K_S -based

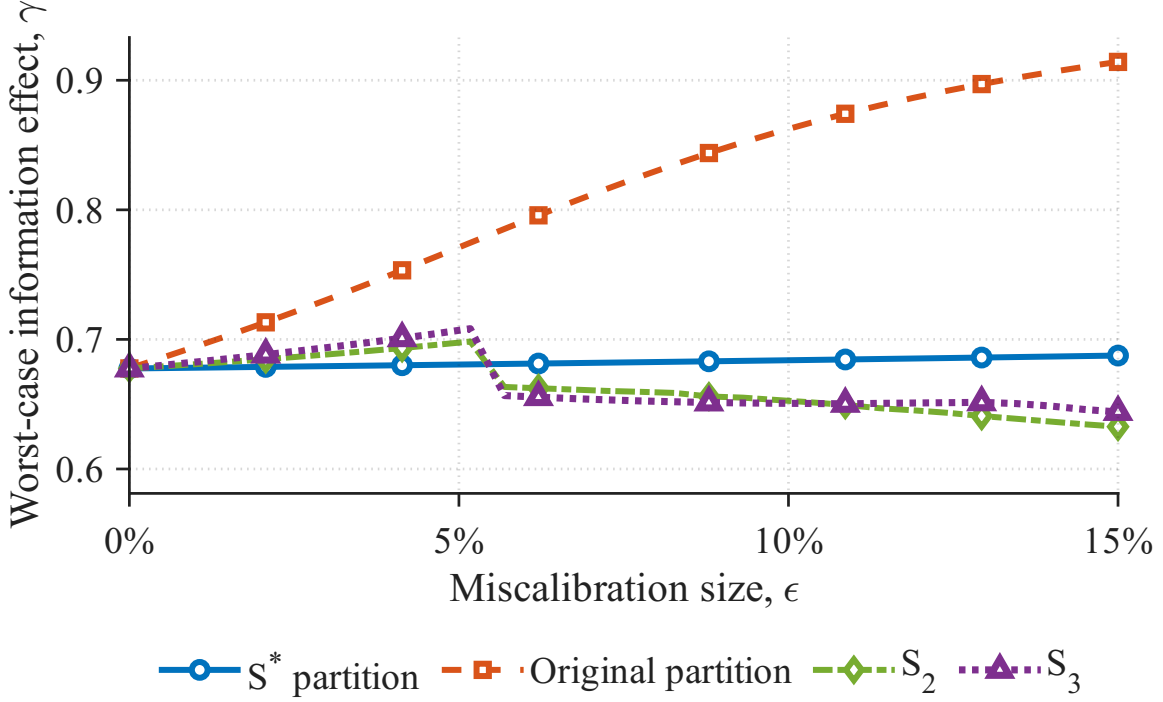


FIGURE 4. Information effect under worst-case miscalibration over different sizes.

ranking, illustrating how sensitivity translates into divergence in worst-case outcomes.

The Monte Carlo results above use the baseline admissible set generated by $\tau_n = (\log n/n)^{1/2}$. Appendix E.1 repeats the ranking and the worst-case Monte Carlo comparison under alternative rank thresholds. The results are substantively stable: S^* remains the top-ranked partition according to K_S for every cut-off, and the finite-sample MSE patterns continue to support the main conclusion that choosing the least-sensitive partition sharply reduces calibration risk.

8. CONCLUSION

This paper studies a common but under-formalized choice in structural estimation: which parameters should be estimated and which should be calibrated. The main contribution is to treat this choice as a partition-selection problem. For each admissible calibration–estimation partition, the paper constructs a scalar

sensitivity statistic that measures the worst-case local effect of calibration errors on the object of interest. The selected partition is the one that minimizes this statistic, and therefore minimizes worst-case local bias induced by plausible miscalibration of the fixed parameters.

The strength of the approach is that it turns sensitivity analysis into a decision rule. Rather than reporting that some calibrated parameters matter, the method provides a criterion for deciding which parameters should be fixed and which should be disciplined by the data. The rule is easy to implement, requires only local derivative information, avoids repeated re-estimation across partitions, and applies to a broad class of structural models. The applications show that this choice can matter substantially: partitions that look similar from the perspective of identification or computational convenience can imply very different robustness properties for the same target object.

The analysis also clarifies the limits of the method. The criterion is local: it evaluates robustness around a reference parameter vector and is informative when calibration errors are plausibly small or moderate. If the reference point is far from the relevant population value, or if large calibration errors are empirically plausible, global sensitivity analysis is more appropriate. A second limitation is that the normalization matrix is part of the economic content of the exercise. It defines what counts as a plausible and comparable miscalibration across parameters. Different normalizations can therefore lead to different selected partitions.

These limitations are not defects of the approach, but conditions for its interpretation. Structural work with calibrated parameters already relies on the assumption that the fixed values are credible approximations. The proposed statistic makes this assumption explicit and forces the researcher to state the scale on which calibration errors are judged. Robustness checks over alternative normalizations, together with the contribution decomposition of the sensitivity statistic, provide a transparent way to assess whether the selected partition depends on fragile choices about plausible parameter ranges.

Overall, the paper's message is that calibration is not merely a computational convenience. It is an econometric choice with direct consequences for the credibility of structural conclusions. By choosing the calibration–estimation partition

through the lens of worst-case local bias, researchers can make this choice more systematic, transparent, and aligned with the object of interest.

APPENDIX A: FURTHER DISCUSSIONS

A.1 Framework and notation for the appendix

This appendix uses the notation of the main text. Unless otherwise stated, all local objects are evaluated at the reference point η^* . For a partition S , write $\eta = (\alpha_S, \beta_S)$, where α_S collects the estimated coordinates and β_S collects the fixed coordinates. When no confusion can arise, the subscript S is suppressed and we write (α, β) . The Jacobian blocks are

$$G_{\alpha,S} := \frac{\partial g(\eta^*; S)}{\partial \alpha'}, \quad G_{\beta,S} := \frac{\partial g(\eta^*; S)}{\partial \beta'},$$

and

$$\Gamma_{\alpha,S} := \frac{\partial \Gamma(\eta^*)}{\partial \alpha'}, \quad \Gamma_{\beta,S} := \frac{\partial \Gamma(\eta^*)}{\partial \beta'}.$$

For any partition satisfying the rank condition in the main text, define

$$D_{\alpha\beta,S} := -(G'_{\alpha,S} W G_{\alpha,S})^{-1} G'_{\alpha,S} W G_{\beta,S}, \quad D_{\gamma\beta,S} := \Gamma_{\alpha,S} D_{\alpha\beta,S} + \Gamma_{\beta,S}. \quad (23)$$

We omit the S subscript when the context is clear.

A.2 Local derivative and first-order expansion

The next lemma records the only differentiability fact needed repeatedly in the appendix and is used in the proof of Lemma 2.

LEMMA 5 (Local derivative and uniform first-order expansion). *Fix $S \in \mathcal{S}_{\text{AD}}(\eta^*)$. Under Assumptions 1 and 2, the following statements hold.*

- (i) $\alpha^* = \alpha(\beta^*; S)$.
- (ii) $\partial \alpha(\beta^*; S) / \partial \beta' = D_{\alpha\beta,S}$ and $\partial \gamma(\beta^*; S) / \partial \beta' = D_{\gamma\beta,S}$, where the matrices are defined in (23).
- (iii) If

$$R_S(\beta) := \gamma(\beta; S) - \gamma(\beta^*; S) - D_{\gamma\beta,S}(\beta - \beta^*),$$

then

$$\sup_{0 < \|\beta - \beta^*\|_2 \leq \delta} \frac{\|R_S(\beta)\|_2}{\|\beta - \beta^*\|_2} \longrightarrow 0 \quad \text{as } \delta \downarrow 0.$$

In particular, for

$$B_\epsilon(\beta^*; S) := \left\{ \beta_0 \in B : \|\Sigma_S^{-1}(\beta^* - \beta_0)\|_2 \leq \epsilon \sqrt{|S^c|} \right\},$$

one has

$$\sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|R_S(\beta_0)\|_2 = o(\epsilon).$$

PROOF. See Appendix C. □

A.3 Using the least-sensitive partition

Let $S^* := S^*(\eta^*)$. After selecting S^* , a researcher may use a nearby alternative calibration $\tilde{\beta}$, re-estimate the free parameters under the same partition, and form

$$\tilde{\eta} := (\alpha(\tilde{\beta}; S^*), \tilde{\beta}).$$

The relevant question is whether the least-sensitive partition remains the same when the criterion is re-evaluated at $\tilde{\eta}$.

ASSUMPTION 5 (Local stability of the admissible set and sensitivity ranking). There exists a neighborhood U_0 of η^* such that $\mathcal{S}_{\text{AD}}(\eta) = \mathcal{S}_{\text{AD}}(\eta^*)$ for all $\eta \in U_0$. Moreover, for every $S \in \mathcal{S}_{\text{AD}}(\eta^*)$, the map $\eta \mapsto K_S(\eta)$ is continuous in a neighborhood of η^* .

THEOREM 6 (Local stability of the selected partition). *Let Assumptions 4 and 5 hold. Then there exists a neighborhood U of η^* such that, for all $\eta \in U$,*

$$S^*(\eta) = S^*(\eta^*).$$

PROOF. See Appendix C. □

In practice, the maintained requirement is that $\tilde{\eta} = (\alpha(\tilde{\beta}; S^*), \tilde{\beta})$ lies in the neighborhood U . This can be checked by recomputing the least-sensitive partition at $\tilde{\eta}$.

A.4 Weak identification and the admissible set of partitions

This subsection gives the local-sequence argument underlying the variance-decay restriction in the definition of $\mathcal{S}_{\text{AD}}(\eta^*)$. Let

$$g_n : \mathcal{A}_S \times \mathcal{B}_S \rightarrow \mathbb{R}^{d_g}$$

denote the population system along the n th element of a local population sequence, and let

$$\eta_n^* := (\alpha_n^*, \beta_n^*)$$

satisfy

$$g_n(\alpha_n^*, \beta_n^*) = 0.$$

The fixed-population case in the main text is obtained by setting $g_n \equiv g$, $\alpha_n^* \equiv \alpha^*$, and $\beta_n^* \equiv \beta^*$.

Throughout this subsection, the fixed component is held at its population reference value β_n^* . Thus, the stochastic object studied here is the conditional minimum-distance estimator at fixed β_n^* , which is the object entering the variance-decay admissibility condition. The use of a sample reference point in plug-in derivative computations is handled separately.

Define

$$\hat{\alpha}_n \equiv \hat{\alpha}_n(\beta_n^*) \in \arg \min_{\alpha \in \mathcal{A}_S} \hat{g}_n(\alpha, \beta_n^*)' \widehat{W} \hat{g}_n(\alpha, \beta_n^*).$$

Equivalently, the sample point studied in this subsection is

$$\hat{\eta}_n^* := (\hat{\alpha}_n(\beta_n^*), \beta_n^*).$$

For $\alpha \in \mathcal{A}_S$, define the sample equation error at the fixed population component by

$$e_n(\alpha, \beta_n^*) := \hat{g}_n(\alpha, \beta_n^*) - g_n(\alpha, \beta_n^*).$$

Let

$$G_n := \frac{\partial g_n(\alpha_n^*, \beta_n^*)}{\partial \alpha'}, \quad C_n := \left(W_n^{1/2} G_n \right)^+.$$

ASSUMPTION 6 (Local-sequence regularity at fixed component). There exist finite constants $K < \infty$, $\bar{r} > 0$, $L_\Gamma < \infty$, $\rho \in [0, 1)$, and a sequence $q_n \rightarrow 0$ such that the following conditions hold for all large n .

(i) $W_n^{1/2}G_n$ has full column rank,

$$C_n W_n^{1/2} G_n = I,$$

and

$$\|W_n^{1/2}\| \leq K.$$

(ii) If $\widehat{W}^{1/2}$ is the sample weighting matrix, then, with probability one,

$$K^{-1}\|W_n^{1/2}v\| \leq \|\widehat{W}^{1/2}v\| \leq K\|W_n^{1/2}v\|$$

for every conformable vector v .

(iii) The estimator is localized. For some $r_n \in (0, \bar{r}]$,

$$P\{\widehat{\alpha}_n(\beta_n^*) \in \mathcal{N}_n(r_n)\} = 1, \quad \mathcal{N}_n(r) := \{\alpha : \|\alpha - \alpha_n^*\| \leq r\}.$$

The map

$$\alpha \mapsto g_n(\alpha, \beta_n^*)$$

is continuously differentiable on $\mathcal{N}_n(\bar{r})$.

(iv) The sample equation satisfies

$$\mathbb{E} [\|e_n(\alpha_n^*, \beta_n^*)\|^2] = O(q_n^2),$$

and

$$\mathbb{E} \left[\sup_{\alpha \in \mathcal{N}_n(\bar{r})} \|e_n(\alpha, \beta_n^*) - e_n(\alpha_n^*, \beta_n^*)\|^2 \right] = O(q_n^2).$$

(v) The Jacobian satisfies

$$\sup_{\alpha \in \mathcal{N}_n(r_n)} \left\| C_n W_n^{1/2} \left[\frac{\partial g_n(\alpha, \beta_n^*)}{\partial \alpha'} - G_n \right] \right\| \leq \rho.$$

(vi) The target is uniformly locally Lipschitz on a neighborhood of the population reference point. For some $\bar{s} > 0$, define

$$\mathcal{B}_n(s) := \{\beta : \|\beta - \beta_n^*\| \leq s\}.$$

The map

$$(\alpha, \beta) \mapsto \Gamma(\alpha, \beta)$$

is continuously differentiable on

$$\mathcal{N}_n(\bar{r}) \times \mathcal{B}_n(\bar{s}),$$

and

$$\sup_{(\alpha, \beta) \in \mathcal{N}_n(\bar{r}) \times \mathcal{B}_n(\bar{s})} \left\| \left[\frac{\partial \Gamma(\alpha, \beta)}{\partial \alpha'} \quad \frac{\partial \Gamma(\alpha, \beta)}{\partial \beta'} \right] \right\| \leq L_\Gamma.$$

PROPOSITION 1 (Second-moment bound for the estimated parameters). Under Assumption 6,

$$\mathbb{E} [\|\hat{\alpha}_n(\beta_n^*) - \alpha_n^*\|^2] = O(\|C_n\|^2 q_n^2).$$

If $\|C_n\|q_n = o(1)$, then

$$\mathbb{E} [\|\hat{\alpha}_n(\beta_n^*) - \alpha_n^*\|^2] = o(1).$$

PROPOSITION 2 (Vanishing second moments for the object of interest). Under Assumption 6, define

$$\hat{\gamma}_n := \Gamma(\hat{\alpha}_n(\beta_n^*), \beta_n^*), \quad \gamma_n^* := \Gamma(\alpha_n^*, \beta_n^*).$$

Then

$$\mathbb{E} [\|\hat{\gamma}_n - \gamma_n^*\|^2] = O(\|C_n\|^2 q_n^2), \quad \text{tr}\{\text{Var}(\hat{\gamma}_n)\} = O(\|C_n\|^2 q_n^2).$$

If $\|C_n\|q_n = o(1)$, then

$$\mathbb{E} [\|\hat{\gamma}_n - \gamma_n^*\|^2] = o(1), \quad \text{tr}\{\text{Var}(\hat{\gamma}_n)\} = o(1).$$

A.5 Consistency of the estimated least-sensitive partition

ASSUMPTION 7 (High-level consistency conditions). The following conditions hold.

- (i) Given the first-stage calibration (β_1, S_1) , $\hat{\eta} \xrightarrow{p} \eta^*$.
- (ii) The estimated admissible set is consistent:

$$\Pr \left\{ \widehat{\mathcal{S}}_{\text{AD}}(\hat{\eta}) = \mathcal{S}_{\text{AD}}(\eta^*) \right\} \rightarrow 1.$$

- (iii) For every candidate partition S ,

$$\sup_{\eta \in N} \left\| \frac{\partial \widehat{g}(\eta; S)}{\partial \eta'} - \frac{\partial g(\eta; S)}{\partial \eta'} \right\| \xrightarrow{p} 0.$$

- (iv) $\widehat{W} \xrightarrow{p} W$, where W is symmetric positive definite.

Assumption 7(iii) is a higher-order uniform convergence condition for the Jacobians used in both the sensitivity statistic and the admissible-set computation. For moment functions $g(\eta; S) = E[m(X_i, \eta; S)]$, primitive sufficient conditions follow from Lemma 2.4 of [Newey and McFadden \(1994\)](#): compactness of N , stationarity and ergodicity of $\{X_i\}$, continuity in η of the derivative of m , and an integrable dominating function for that derivative.

THEOREM 7 (Consistency of the selected partition). *Suppose Assumptions 1, 2, 4, and 7 hold. Then*

$$\Pr \left\{ \widehat{S}^*(\hat{\eta}) = S^*(\eta^*) \right\} \rightarrow 1.$$

PROOF. See Appendix C. □

A.6 Direct and re-estimated derivatives

This subsection formalizes why Algorithm in Section 5 can compute the local derivative at the common reference point rather than re-estimating the model separately for every candidate partition. Fix a partition S . For a generic fixed value β , define

$$\widehat{\alpha}(\beta; S) \in \arg \min_{\alpha \in A_S} \widehat{g}(\alpha, \beta; S)' \widehat{W} \widehat{g}(\alpha, \beta; S).$$

Let

$$\widehat{G}_\alpha(\eta; S) := \frac{\partial \widehat{g}(\eta; S)}{\partial \alpha'}, \quad \widehat{G}_\beta(\eta; S) := \frac{\partial \widehat{g}(\eta; S)}{\partial \beta'}.$$

For the S -specific ordering of the common reference point, write $\widehat{\eta} = (\widehat{\alpha}_S, \widehat{\beta}_S)$. The direct derivative used by the algorithm is

$$\widehat{D}_{\alpha\beta}(S) := - \left[\widehat{G}_\alpha(\widehat{\eta}; S)' \widehat{W} \widehat{G}_\alpha(\widehat{\eta}; S) \right]^{-1} \widehat{G}_\alpha(\widehat{\eta}; S)' \widehat{W} \widehat{G}_\beta(\widehat{\eta}; S). \quad (24)$$

ASSUMPTION 8 (Local regularity for direct derivative computation). For the fixed partition S , let $\eta^* = (\alpha^*, \beta^*)$ denote the population reference point under the S -specific ordering. The following conditions hold.

- (i) $\eta^* \in \text{int}(N)$, $g(\eta^*; S) = 0$, and $G_{\alpha,S}$ has full column rank.
- (ii) $\widehat{\eta} \xrightarrow{p} \eta^*$ and $\widehat{\beta}_S \xrightarrow{p} \beta^*$.
- (iii) $\widehat{W} \xrightarrow{p} W$, where W is symmetric positive definite.
- (iv) The local sample solution $\widehat{\alpha}(\widehat{\beta}_S; S)$ exists with probability approaching one, satisfies the sample first-order conditions, and

$$(\widehat{\alpha}(\widehat{\beta}_S; S), \widehat{\beta}_S) \xrightarrow{p} (\alpha^*, \beta^*).$$

- (v) In a neighborhood of η^* , $\widehat{g}$, its first derivatives, and its second derivatives converge uniformly in probability to their population counterparts.

THEOREM 8 (Equivalence of direct and re-estimated derivatives). *Suppose Assumption 8 holds. Then, with probability approaching one, the derivative of the sample re-estimation map exists at $\widehat{\beta}_S$, and*

$$\left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\widehat{\beta}_S} = \widehat{D}_{\alpha\beta}(S) + o_p(1) = D_{\alpha\beta,S} + o_p(1),$$

where $D_{\alpha\beta,S}$ is defined in (23).

PROOF. See Appendix C. □

COROLLARY 1 (Derivative of the object of interest). Suppose the conditions of Theorem 8 hold and Γ is continuously differentiable in a neighborhood of η^* .

Define

$$\widehat{D}_{\gamma\beta}(S) := \Gamma_{\alpha}(\widehat{\eta}; S) \widehat{D}_{\alpha\beta}(S) + \Gamma_{\beta}(\widehat{\eta}; S).$$

Then

$$\left. \frac{\partial \widehat{\gamma}(\beta; S)}{\partial \beta'} \right|_{\beta=\widehat{\beta}_S} = \widehat{D}_{\gamma\beta}(S) + o_p(1) = D_{\gamma\beta, S} + o_p(1).$$

PROOF. See Appendix C. □

REMARK 4 (Uniformity over partitions). Let $\mathcal{S}_0$ be a finite collection of candidate partitions. If Assumption 8 holds uniformly over $S \in \mathcal{S}_0$ and

$$\inf_{S \in \mathcal{S}_0} \lambda_{\min}(G'_{\alpha, S} W G_{\alpha, S}) > 0,$$

then

$$\max_{S \in \mathcal{S}_0} \left\| \left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\widehat{\beta}_S} - \widehat{D}_{\alpha\beta}(S) \right\| = o_p(1),$$

and the analogous statement holds for $\widehat{D}_{\gamma\beta}(S)$. Hence the direct algorithmic derivative is uniformly asymptotically equivalent to the derivative obtained after partition-specific re-estimation.

APPENDIX B: EXTENSION TO MLE

The examples below clarify when the population system $g(\eta; S)$ depends on the partition.

EXAMPLE (Maximum likelihood). In maximum likelihood estimation, the population moments are score equations. For a partition S ,

$$g_P(\alpha, \beta; S) := E_P \left[\frac{\partial}{\partial \alpha} \log f(X_i | \alpha, \beta) \right]. \quad (25)$$

Here S selects the score components included in the system. Hence both the functional form and the dimension $d_g(S)$ of $g_P(\cdot; S)$ may depend on S .

EXAMPLE (Classical minimum distance). In the classical minimum-distance framework,

$$g_P(\alpha, \beta; S) = \theta(P) - \psi(\alpha, \beta). \quad (26)$$

The same population restrictions are used for all partitions, up to the relabeling of coordinates into estimated and fixed blocks.

In CMD, and in GMM settings where the moment conditions do not depend on the partition, a primitive condition for Assumption 2 to hold is that the first-stage calibration is compatible with the population equations, i.e., there exists α such that $g(\alpha, \beta_1; S_1) = 0$.

In MLE, by contrast, $g_P(\eta; S)$ typically depends on S because S selects a subset of score equations. A convenient sufficient condition for Assumption 2 is that η^* satisfies the mean-zero restriction for the full score vector, not only for the score components associated with the first-stage estimated parameters. Appendix B.1 constructs such a reference point.

B.1 Reference point for the MLE case

Suppose the researcher first estimates the model by MLE under a calibration strategy (S_1, β_1) and obtains

$$\tilde{\eta}_n := (\hat{\alpha}_n(\beta_1; S_1), \beta_1).$$

As discussed above, the MLE moment map $g_P(\eta; S)$ depends on S . Consequently, $\tilde{\eta}_n$ need not satisfy the mean-zero restrictions for score components corresponding to parameters that were fixed in the first stage. This may violate Assumption 2 when the reference point must be observationally equivalent across candidate partitions.

Let

$$S_{\text{full}} := \{1, \dots, d_\eta\}, \quad s(X_i, \eta) := \frac{\partial}{\partial \eta} \log f(X_i | \eta),$$

and define the full-score sample and population moments by

$$\widehat{g}_n(\eta; S_{\text{full}}) := \frac{1}{n} \sum_{i=1}^n s(X_i, \eta), \quad g(\eta; S_{\text{full}}) := E_P[s(X_i, \eta)].$$

Let

$$Q_n(\eta) := \widehat{g}_n(\eta; S_{\text{full}})' W_n \widehat{g}_n(\eta; S_{\text{full}}), \quad Q(\eta) := g(\eta; S_{\text{full}})' W g(\eta; S_{\text{full}}),$$

where $W_n \xrightarrow{p} W$ and W is symmetric positive definite. To avoid confusion with the weighting matrix, let H denote a fixed symmetric positive-definite penalty metric and write $\|x\|_H^2 := x' H x$. Define

$$\widehat{\eta}_n \in \arg \min_{\eta \in N} \{Q_n(\eta) + \lambda_n \|\eta - \widetilde{\eta}_n\|_H^2\}. \quad (27)$$

The special case $H = I_{d_\eta}$ gives the Euclidean penalty.

Define the full-score identified set

$$N_0(S_{\text{full}}) := \{\eta \in N : g(\eta; S_{\text{full}}) = 0\}.$$

Since every partition-specific score system is a subvector of the full score, any $\eta \in N_0(S_{\text{full}})$ also belongs to $N_0(S)$ for every partition S . If $\widetilde{\eta}_n \xrightarrow{p} \widetilde{\eta}_0$, define

$$\eta^* := \arg \min_{\eta \in N_0(S_{\text{full}})} \|\eta - \widetilde{\eta}_0\|_H^2. \quad (28)$$

Thus η^* is the element of the full-score identified set closest to the probability limit of the first-stage estimate.

ASSUMPTION 9 (Penalized full-score reference point). The following conditions hold.

- (i) $N \subset \mathbb{R}^{d_\eta}$ is compact and H is symmetric positive definite.
- (ii) $\widetilde{\eta}_n \xrightarrow{p} \widetilde{\eta}_0$.
- (iii) $\sup_{\eta \in N} |Q_n(\eta) - Q(\eta)| \xrightarrow{p} 0$.
- (iv) $N_0(S_{\text{full}})$ is nonempty and separated by Q : for every $\varepsilon > 0$,

$$\inf_{\eta \in N : d_H(\eta, N_0(S_{\text{full}})) \geq \varepsilon} Q(\eta) > 0, \quad d_H(\eta, N_0) := \inf_{\zeta \in N_0} \|\eta - \zeta\|_H.$$

- (v) The projection in (28) exists and is unique.
- (vi) Along the full-score identified set,

$$\sup_{\eta \in N_0(S_{\text{full}})} Q_n(\eta) = O_p(n^{-1}).$$

- (vii) The tuning parameter satisfies $\lambda_n \rightarrow 0$ and $n\lambda_n \rightarrow \infty$.

Assumption 9 collects the conditions needed for the penalized full-score estimator. Parts (i)–(iii) are standard extremum-estimation conditions. Parts (iv)–(vi) handle the possibility that the full-score problem is set-identified. Part (vii) ensures that the penalty vanishes asymptotically but dominates the $O_p(n^{-1})$ stochastic variation of Q_n along flat directions. A convenient deterministic choice is $\lambda_n = n^{-\kappa}$ with $0 < \kappa < 1$. If one also wants the penalty to be negligible at the $\sqrt{n}$ scale in well-identified directions, one may choose $\kappa \in (1/2, 1)$, for instance $\lambda_n = n^{-2/3}$.

THEOREM 9 (Consistency of the penalized full-score reference point). *Under Assumption 9,*

$$\hat{\eta}_n \xrightarrow{p} \eta^*.$$

Consequently, if $\eta^ \in \text{int}(N)$, the estimator in (27) delivers a reference point satisfying Assumption 2 for the MLE partition-selection problem.*

PROOF. See Appendix C. □

APPENDIX C: MAIN PROOFS

PROOF OF LEMMA 5. Fix a partition $S \in \mathcal{S}_{\text{AD}}(\eta^*)$. Under the S -specific ordering, write $\eta^* = (\alpha^*, \beta^*)$. By Assumption 2, the reference point satisfies $g(\alpha^*, \beta^*; S) = 0$. The minimum-distance criterion at the fixed value β^* is

$$Q(\alpha, \beta^*; S) := g(\alpha, \beta^*; S)' W g(\alpha, \beta^*; S).$$

Since W is positive definite, $Q(\alpha, \beta^*; S) \geq 0$ for every α . At $\alpha = \alpha^*$, the criterion equals zero. Hence α^* solves the conditional MD problem at β^* , that is,

$$\alpha^* = \alpha(\beta^*; S).$$

This proves part (i).

To obtain the derivative, define the population first-order condition map

$$F(\alpha, \beta; S) := \frac{\partial Q(\alpha, \beta; S)}{\partial \alpha} = 2G_\alpha(\alpha, \beta; S)' W g(\alpha, \beta; S) \in \mathbb{R}^{|S|},$$

where

$$G_\alpha(\alpha, \beta; S) := \frac{\partial g(\alpha, \beta; S)}{\partial \alpha'}.$$

By Assumption 1, g is twice continuously differentiable, so F is continuously differentiable in a neighborhood of (α^*, β^*) . Since $g(\alpha^*, \beta^*; S) = 0$,

$$F(\alpha^*, \beta^*; S) = 0.$$

The derivative of F with respect to α , evaluated at the reference point, is

$$\frac{\partial F(\alpha^*, \beta^*; S)}{\partial \alpha'} = 2G'_{\alpha, S} W G_{\alpha, S},$$

because all terms involving second derivatives of g are multiplied by $g(\alpha^*, \beta^*; S) = 0$. Similarly,

$$\frac{\partial F(\alpha^*, \beta^*; S)}{\partial \beta'} = 2G'_{\alpha, S} W G_{\beta, S}.$$

The rank condition in $S \in \mathcal{S}_{AD}(\eta^*)$ gives $\text{rank}(G_{\alpha, S}) = |S|$. Since W is positive definite, $G'_{\alpha, S} W G_{\alpha, S}$ is nonsingular. Therefore the implicit function theorem applied to $F(\alpha, \beta; S) = 0$ yields a neighborhood of β^* and a unique differentiable map $\beta \mapsto \alpha(\beta; S)$ satisfying

$$F(\alpha(\beta; S), \beta; S) = 0, \quad \alpha(\beta^*; S) = \alpha^*.$$

Differentiating the identity $F(\alpha(\beta; S), \beta; S) = 0$ at $\beta = \beta^*$ gives

$$0 = \frac{\partial F}{\partial \alpha'} \frac{\partial \alpha(\beta^*; S)}{\partial \beta'} + \frac{\partial F}{\partial \beta'}.$$

Substituting the two derivative blocks above and cancelling the common factor 2,

$$\frac{\partial \alpha(\beta^*; S)}{\partial \beta'} = - (G'_{\alpha, S} W G_{\alpha, S})^{-1} G'_{\alpha, S} W G_{\beta, S} = D_{\alpha \beta, S}.$$

Since Γ is continuously differentiable by Assumption 1, the chain rule applied to

$$\gamma(\beta; S) = \Gamma(\alpha(\beta; S), \beta)$$

gives

$$\frac{\partial \gamma(\beta^*; S)}{\partial \beta'} = \Gamma_{\alpha, S} D_{\alpha \beta, S} + \Gamma_{\beta, S} = D_{\gamma \beta, S}.$$

This proves part (ii).

It remains to prove the first-order expansion. Since $\gamma(\cdot; S)$ is differentiable at β^* , it is Fréchet differentiable in the finite-dimensional Euclidean space, and therefore

$$\frac{\|\gamma(\beta; S) - \gamma(\beta^*; S) - D_{\gamma \beta, S}(\beta - \beta^*)\|_2}{\|\beta - \beta^*\|_2} \longrightarrow 0 \quad \text{as } \beta \rightarrow \beta^*, \beta \neq \beta^*.$$

With the notation

$$R_S(\beta) := \gamma(\beta; S) - \gamma(\beta^*; S) - D_{\gamma \beta, S}(\beta - \beta^*),$$

this is exactly the pointwise statement

$$\frac{\|R_S(\beta)\|_2}{\|\beta - \beta^*\|_2} \rightarrow 0 \quad \text{as } \beta \rightarrow \beta^*.$$

We now turn this pointwise statement into the displayed uniform statement. Suppose, toward a contradiction, that

$$\sup_{0 < \|\beta - \beta^*\|_2 \leq \delta} \frac{\|R_S(\beta)\|_2}{\|\beta - \beta^*\|_2} \not\rightarrow 0 \quad \text{as } \delta \downarrow 0.$$

Then there exist $c > 0$, a sequence $\delta_m \downarrow 0$, and points β_m with $0 < \|\beta_m - \beta^*\|_2 \leq \delta_m$ such that

$$\frac{\|R_S(\beta_m)\|_2}{\|\beta_m - \beta^*\|_2} \geq c.$$

But $\|\beta_m - \beta^*\|_2 \leq \delta_m \rightarrow 0$, so $\beta_m \rightarrow \beta^*$, contradicting the pointwise Fréchet differentiability limit. Hence

$$\sup_{0 < \|\beta - \beta^*\|_2 \leq \delta} \frac{\|R_S(\beta)\|_2}{\|\beta - \beta^*\|_2} \rightarrow 0 \quad \text{as } \delta \downarrow 0.$$

Finally, consider the normalized neighborhood

$$B_\epsilon(\beta^*; S) := \left\{ \beta_0 \in B : \|\Sigma_S^{-1}(\beta^* - \beta_0)\|_2 \leq \epsilon \sqrt{|S^c|} \right\}.$$

For any $\beta_0 \in B_\epsilon(\beta^*; S)$,

$$\|\beta_0 - \beta^*\|_2 = \|\Sigma_S \Sigma_S^{-1}(\beta_0 - \beta^*)\|_2 \leq \|\Sigma_S\|_2 \|\Sigma_S^{-1}(\beta_0 - \beta^*)\|_2 \leq \|\Sigma_S\|_2 \epsilon \sqrt{|S^c|}.$$

Thus $B_\epsilon(\beta^*; S)$ is contained in a Euclidean ball around β^* of radius $C_S \epsilon$, where $C_S := \|\Sigma_S\|_2 \sqrt{|S^c|}$. Applying the uniform differentiability bound on this ball gives

$$\sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|R_S(\beta_0)\|_2 \leq \left(\sup_{0 < \|\beta - \beta^*\|_2 \leq C_S \epsilon} \frac{\|R_S(\beta)\|_2}{\|\beta - \beta^*\|_2} \right) C_S \epsilon = o(\epsilon).$$

This proves part (iii) and completes the proof of the lemma. $\square$

PROOF OF THEOREM 9. Let

$$P_n(\eta) := \|\eta - \tilde{\eta}_n\|_H^2, \quad P(\eta) := \|\eta - \tilde{\eta}_0\|_H^2.$$

By definition of $\hat{\eta}_n$ as a minimizer of (27), and because $\eta^* \in N_0(S_{\text{full}})$, we have

$$Q_n(\hat{\eta}_n) + \lambda_n P_n(\hat{\eta}_n) \leq Q_n(\eta^*) + \lambda_n P_n(\eta^*). \quad (29)$$

By Assumption 9(vi), since $\eta^* \in N_0(S_{\text{full}})$,

$$Q_n(\eta^*) = O_p(n^{-1}).$$

Combining this with Assumption 9(vii), which implies $n\lambda_n \rightarrow \infty$, gives

$$\frac{Q_n(\eta^*)}{\lambda_n} = o_p(1).$$

Using (29), we therefore obtain

$$Q_n(\hat{\eta}_n) \leq Q_n(\eta^*) + \lambda_n P_n(\eta^*).$$

The first term is $O_p(n^{-1}) = o_p(1)$. The second term is also $o_p(1)$, because $\lambda_n \rightarrow 0$ by Assumption 9(vii) and $P_n(\eta^*) = O_p(1)$ by compactness of N and Assumption 9(ii). Hence

$$Q_n(\hat{\eta}_n) = o_p(1).$$

Next, Assumption 9(iii) gives uniform convergence of Q_n to Q , so

$$Q(\hat{\eta}_n) \leq Q_n(\hat{\eta}_n) + \sup_{\eta \in N} |Q_n(\eta) - Q(\eta)| = o_p(1).$$

By the separation condition in Assumption 9(iv), this implies

$$d_H(\hat{\eta}_n, N_0(S_{\text{full}})) \xrightarrow{p} 0. \quad (30)$$

Indeed, if the distance from $\hat{\eta}_n$ to $N_0(S_{\text{full}})$ were bounded away from zero with non-vanishing probability, the population criterion Q would be bounded away from zero on that event, contradicting $Q(\hat{\eta}_n) = o_p(1)$.

Return now to (29) and divide both sides by λ_n :

$$P_n(\hat{\eta}_n) \leq P_n(\eta^*) + \frac{Q_n(\eta^*)}{\lambda_n} = P_n(\eta^*) + o_p(1).$$

Since $\tilde{\eta}_n \xrightarrow{p} \tilde{\eta}_0$ and N is compact,

$$\sup_{\eta \in N} |P_n(\eta) - P(\eta)| \xrightarrow{p} 0.$$

To see this, note that

$$|P_n(\eta) - P(\eta)| = \left| \|\eta - \tilde{\eta}_n\|_H^2 - \|\eta - \tilde{\eta}_0\|_H^2 \right|$$

is uniformly controlled over compact N by a constant times $\|\tilde{\eta}_n - \tilde{\eta}_0\|_H$. Consequently,

$$P(\hat{\eta}_n) \leq P(\eta^*) + o_p(1). \quad (31)$$

We now use a subsequence argument. Take any subsequence of $\{\hat{\eta}_n\}$. By compactness of N , it has a further subsequence, still denoted by $\{\hat{\eta}_n\}$ for notational simplicity, that converges in distribution to some limit $\hat{\eta}$. By (30), every such limit satisfies

$$\hat{\eta} \in N_0(S_{\text{full}}) \quad \text{a.s.}$$

Moreover, by continuity of P and (31),

$$P(\hat{\eta}) \leq P(\eta^*) \quad \text{a.s.}$$

But Assumption 9(v) states that η^* is the unique minimizer of P over $N_0(S_{\text{full}})$. Therefore

$$\hat{\eta} = \eta^* \quad \text{a.s.}$$

Since every subsequence has a further subsequence converging to η^* , it follows that

$$\hat{\eta}_n \xrightarrow{p} \eta^*.$$

This proves the theorem. □

PROOF OF LEMMA 2. Fix a partition $S \in \mathcal{S}_{\text{AD}}(\eta^*)$ if $|S^c| = 0$, there are no fixed parameters, the sensitivity statistic $K_S = 0$, and the worst-case bias is exactly zero, so the equality holds trivially. Assume hereafter that $|S^c| > 0$. Write

$$D := \frac{\partial \gamma(\beta^*; S)}{\partial \beta'} = D_{\gamma\beta, S}.$$

By Lemma 5, for $\beta_0 \in B_\epsilon(\beta^*; S)$ we have the expansion

$$\gamma(\beta_0; S) - \gamma(\beta^*; S) = D(\beta_0 - \beta^*) + R_S(\beta_0), \quad \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|R_S(\beta_0)\|_2 = o(\epsilon).$$

Equivalently,

$$\gamma(\beta^*; S) - \gamma(\beta_0; S) = D(\beta^* - \beta_0) - R_S(\beta_0).$$

By the triangle inequality, for each $\beta_0 \in B_\epsilon(\beta^*; S)$,

$$\|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 \leq \|D(\beta^* - \beta_0)\|_2 + \|R_S(\beta_0)\|_2.$$

Taking the supremum over $B_\epsilon(\beta^*; S)$ gives

$$\sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 \leq \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|D(\beta^* - \beta_0)\|_2 + \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|R_S(\beta_0)\|_2. \quad (32)$$

Conversely, using

$$D(\beta^* - \beta_0) = \gamma(\beta^*; S) - \gamma(\beta_0; S) + R_S(\beta_0),$$

the triangle inequality gives, for each β_0 ,

$$\|D(\beta^* - \beta_0)\|_2 \leq \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 + \|R_S(\beta_0)\|_2.$$

Taking the supremum gives

$$\sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|D(\beta^* - \beta_0)\|_2 \leq \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 + \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|R_S(\beta_0)\|_2. \quad (33)$$

Combining (32) and (33),

$$\left| \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 - \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|D(\beta^* - \beta_0)\|_2 \right| \leq \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|R_S(\beta_0)\|_2 = o(\epsilon). \quad (34)$$

Thus,

$$\text{WorstBias}(\beta^*, \epsilon; S) := \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 = \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|D(\beta^* - \beta_0)\|_2 + o(\epsilon).$$

Now reparametrize the constraint. For $\beta_0 \in B_\epsilon(\beta^*; S)$, define

$$x := \frac{1}{\epsilon \sqrt{|S^c|}} \Sigma_S^{-1}(\beta^* - \beta_0).$$

Then $\|x\|_2 \leq 1$ and

$$\beta^* - \beta_0 = \epsilon \sqrt{|S^c|} \Sigma_S x.$$

Therefore

$$\begin{aligned} \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \|D(\beta^* - \beta_0)\|_2 &= \epsilon \sqrt{|S^c|} \sup_{\|x\|_2 \leq 1} \|D \Sigma_S x\|_2 \\ &= \epsilon \sqrt{|S^c|} \|D \Sigma_S\|_2, \end{aligned}$$

where the last equality is the definition of the induced Euclidean operator norm. Recalling

$$K_S := \sqrt{|S^c|} \left\| \frac{\partial \gamma(\beta^*; S)}{\partial \beta'} \Sigma_S \right\|_2 = \sqrt{|S^c|} \|D\Sigma_S\|_2,$$

we conclude that

$$\text{WorstBias}(\beta^*, \epsilon; S) = \epsilon K_S + o(\epsilon).$$

This completes the proof. $\square$

PROOF OF THEOREM 3. Let

$$\mathcal{S}_0 := \mathcal{S}_{\text{AD}}(\eta^*), \quad W_S(\epsilon) := \text{WorstBias}(\beta^*, \epsilon; S).$$

Since $d_\eta < \infty$, the number of possible partitions is finite, and hence $\mathcal{S}_0$ is finite. Let

$$S^* = \arg \min_{S \in \mathcal{S}_0} K_S.$$

By Assumption 4, this minimizer is unique. By Lemma 2, for each $S \in \mathcal{S}_0$,

$$W_S(\epsilon) = \epsilon K_S + o_S(\epsilon).$$

Equivalently, for $0 < \epsilon$ small,

$$W_S(\epsilon) = \epsilon \{K_S + r_S(\epsilon)\}, \quad r_S(\epsilon) \rightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

Because $\mathcal{S}_0$ is finite,

$$\rho(\epsilon) := \max_{S \in \mathcal{S}_0} |r_S(\epsilon)| \longrightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

If $\mathcal{S}_0 = \{S^*\}$, the result is immediate. Otherwise, define the strict gap

$$\Delta := \min_{S \in \mathcal{S}_0, S \neq S^*} \{K_S - K_{S^*}\}.$$

By uniqueness of S^* , $\Delta > 0$. Since $\rho(\epsilon) \rightarrow 0$, there exists $E > 0$ such that, for all $0 < \epsilon \leq E$,

$$\rho(\epsilon) < \frac{\Delta}{3}.$$

Hence, for any $S \in \mathcal{S}_0$ with $S \neq S^*$ and any $0 < \epsilon \leq E$,

$$\begin{aligned}
W_S(\epsilon) - W_{S^*}(\epsilon) &= \epsilon\{K_S - K_{S^*}\} + \epsilon\{r_S(\epsilon) - r_{S^*}(\epsilon)\} \\
&\geq \epsilon\Delta - 2\epsilon\rho(\epsilon) \\
&> \epsilon\Delta - \frac{2}{3}\epsilon\Delta \\
&= \frac{1}{3}\epsilon\Delta > 0.
\end{aligned}$$

Therefore $W_{S^*}(\epsilon) < W_S(\epsilon)$ for every $S \neq S^*$ and every $0 < \epsilon \leq E$. Thus S^* is the unique minimizer of $\text{WorstBias}(\beta^*, \epsilon; S)$ over $\mathcal{S}_0$ for all $0 < \epsilon \leq E$. Since S^* is also the unique minimizer of K_S , we obtain

$$\arg \min_{S \in \mathcal{S}_{\text{AD}}(\eta^*)} K_S = \arg \min_{S \in \mathcal{S}_{\text{AD}}(\eta^*)} \text{WorstBias}(\beta^*, \epsilon; S), \quad 0 < \epsilon \leq E.$$

This proves the theorem. □

PROOF OF THEOREM 4. Let

$$\mathcal{S}_0 := \mathcal{S}_{\text{AD}}(\eta^*), \quad W_S(\epsilon) := \text{WorstBias}(\beta^*, \epsilon; S),$$

and define

$$R_S(n) := \mathbb{E} \left[\|\hat{\gamma}(\beta^*; S) - \gamma(\beta^*; S)\|_2^2 \right].$$

By the definition of the admissible set, every $S \in \mathcal{S}_0$ satisfies the second-moment decay condition

$$R_S(n) = o(1).$$

Since $\mathcal{S}_0$ is finite, this convergence is uniform over admissible partitions. Thus, defining

$$r_n := \max_{S \in \mathcal{S}_0} R_S(n),$$

we have

$$r_n = o(1).$$

For each admissible partition, define the finite-sample mean bias

$$m_S(n) := \mathbb{E}[\hat{\gamma}(\beta^*; S)] - \gamma(\beta^*; S),$$

and let

$$V_S(n) := \text{tr}\{\text{Var}(\hat{\gamma}(\beta^*; S))\}.$$

The second-moment identity gives

$$R_S(n) = V_S(n) + \|m_S(n)\|_2^2.$$

Therefore, uniformly over $S \in \mathcal{S}_0$,

$$V_S(n) \leq R_S(n) \leq r_n = o(1), \quad \|m_S(n)\|_2 \leq R_S(n)^{1/2} \leq r_n^{1/2} = o(1).$$

Now fix $S \in \mathcal{S}_0$ and $\beta_0 \in B_\epsilon(\beta^*; S)$. Write

$$b_S(\beta_0) := \gamma(\beta^*; S) - \gamma(\beta_0; S).$$

Using

$$\hat{\gamma}(\beta^*; S) - \gamma(\beta_0; S) = b_S(\beta_0) + [\hat{\gamma}(\beta^*; S) - \gamma(\beta^*; S)],$$

we obtain the exact decomposition

$$\begin{aligned} \text{MSE}(\beta^*, \beta_0, S) &= \mathbb{E} \left[\|\hat{\gamma}(\beta^*; S) - \gamma(\beta_0; S)\|_2^2 \right] \\ &= \|b_S(\beta_0)\|_2^2 + 2b_S(\beta_0)' m_S(n) + R_S(n). \end{aligned}$$

Hence

$$\begin{aligned} &\left| \text{MSE}(\beta^*, \beta_0, S) - \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2^2 \right| \\ &\leq 2 \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2 \|m_S(n)\|_2 + R_S(n) \\ &\leq 2W_S(\epsilon) r_n^{1/2} + r_n. \end{aligned}$$

For fixed $0 < \epsilon \leq E$, and because $\mathcal{S}_0$ is finite,

$$\bar{W}_\epsilon := \max_{S \in \mathcal{S}_0} W_S(\epsilon) < \infty.$$

Therefore

$$\sup_{S \in \mathcal{S}_0} \sup_{\beta_0 \in B_\epsilon(\beta^*; S)} \left| \text{MSE}(\beta^*, \beta_0, S) - \|\gamma(\beta^*; S) - \gamma(\beta_0; S)\|_2^2 \right| \leq 2\bar{W}_\epsilon r_n^{1/2} + r_n = o(1).$$

It follows that, uniformly over $S \in \mathcal{S}_0$,

$$\max_{\beta_0 \in B_\epsilon(\beta^*; S)} \text{MSE}(\beta^*, \beta_0, S) = W_S(\epsilon)^2 + \Delta_S(n, \epsilon),$$

where

$$\sup_{S \in \mathcal{S}_0} |\Delta_S(n, \epsilon)| = o(1).$$

Therefore,

$$\begin{aligned} & \min_{S \in \mathcal{S}_{\text{AD}}(\eta^*)} \max_{\beta_0 \in B_\epsilon(\beta^*; S)} \text{MSE}(\beta^*, \beta_0, S) \\ &= \min_{S \in \mathcal{S}_0} \{W_S(\epsilon)^2 + \Delta_S(n, \epsilon)\} \\ &= \min_{S \in \mathcal{S}_0} W_S(\epsilon)^2 + o(1). \end{aligned}$$

By Theorem 3, there exists $E > 0$ such that, for every $0 < \epsilon \leq E$, the least-sensitive partition S^* minimizes $W_S(\epsilon)$ over $\mathcal{S}_{\text{AD}}(\eta^*)$. Hence

$$\min_{S \in \mathcal{S}_0} W_S(\epsilon)^2 = W_{S^*}(\epsilon)^2.$$

Since

$$W_{S^*}(\epsilon) = \text{WorstBias}(\beta^*, \epsilon; S^*),$$

we conclude that

$$\min_{S \in \mathcal{S}_{\text{AD}}(\eta^*)} \max_{\beta_0 \in B_\epsilon(\beta^*; S)} \text{MSE}(\beta^*, \beta_0, S) = \text{WorstBias}(\beta^*, \epsilon; S^*)^2 + o(1).$$

This proves the theorem. □

PROOF OF THEOREM 6. Let

$$\mathcal{S}_0 := \mathcal{S}_{\text{AD}}(\eta^*).$$

By Assumption 5, there exists a neighborhood U_0 of η^* such that

$$\mathcal{S}_{\text{AD}}(\eta) = \mathcal{S}_0 \quad \text{for all } \eta \in U_0.$$

Since the number of partitions is finite, $\mathcal{S}_0$ is finite. Let

$$S^* := S^*(\eta^*) \in \arg \min_{S \in \mathcal{S}_0} K_S(\eta^*).$$

By Assumption 4, this minimizer is unique. If $\mathcal{S}_0 = \{S^*\}$, the result is immediate. Otherwise define the strict gap

$$\Delta := \min_{S \in \mathcal{S}_0, S \neq S^*} \{K_S(\eta^*) - K_{S^*}(\eta^*)\}.$$

By uniqueness, $\Delta > 0$.

By Assumption 5, for every $S \in \mathcal{S}_0$, the map $\eta \mapsto K_S(\eta)$ is continuous in a neighborhood of η^* . Because $\mathcal{S}_0$ is finite, there exists a neighborhood $U_1 \subseteq U_0$ of η^* such that, for every $\eta \in U_1$ and every $S \in \mathcal{S}_0$,

$$|K_S(\eta) - K_S(\eta^*)| < \frac{\Delta}{3}.$$

Then, for any $S \in \mathcal{S}_0$ with $S \neq S^*$ and any $\eta \in U_1$,

$$\begin{aligned} K_S(\eta) - K_{S^*}(\eta) &= [K_S(\eta^*) - K_{S^*}(\eta^*)] + [K_S(\eta) - K_S(\eta^*)] - [K_{S^*}(\eta) - K_{S^*}(\eta^*)] \\ &\geq \Delta - \frac{\Delta}{3} - \frac{\Delta}{3} = \frac{\Delta}{3} > 0. \end{aligned}$$

Thus $K_{S^*}(\eta) < K_S(\eta)$ for every $S \in \mathcal{S}_0 \setminus \{S^*\}$ and every $\eta \in U_1$. Since $\mathcal{S}_{\text{AD}}(\eta) = \mathcal{S}_0$ on U_1 , it follows that

$$S^*(\eta) = S^*(\eta^*) \quad \text{for all } \eta \in U_1.$$

Taking $U := U_1$ proves the theorem. □

PROOF OF PROPOSITION 1. Let

$$\Delta_n := \hat{\alpha}_n(\beta_n^*) - \alpha_n^*.$$

For this proof, set

$$A_n := W_n^{1/2}, \quad \widehat{A}_n := \widehat{W}^{1/2}, \quad J_n(\alpha) := \frac{\partial g_n(\alpha, \beta_n^*)}{\partial \alpha'}, \quad G_n := J_n(\alpha_n^*),$$

so that $C_n = (A_n G_n)^+$. All pathwise inequalities below are understood on the probability-one event on which the localization and weight-equivalence conditions in Assumption 6 hold.

Since

$$\widehat{\alpha}_n(\beta_n^*) \in \mathcal{N}_n(r_n)$$

and $\mathcal{N}_n(r_n)$ is convex, the line segment

$$\alpha_n^* + t\Delta_n, \quad t \in [0, 1],$$

lies inside $\mathcal{N}_n(r_n)$. A first-order Taylor expansion of $\alpha \mapsto g_n(\alpha, \beta_n^*)$ around α_n^* gives

$$g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*) = g_n(\alpha_n^*, \beta_n^*) + G_n \Delta_n + R_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*), \quad (35)$$

where the Taylor residual has the mean-value integral form

$$R_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*) = \int_0^1 [J_n(\alpha_n^* + t\Delta_n) - G_n] \Delta_n dt. \quad (36)$$

Since

$$g_n(\alpha_n^*, \beta_n^*) = 0,$$

equation (35) becomes

$$g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*) = G_n \Delta_n + R_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*).$$

Premultiplying by A_n and then by C_n gives

$$C_n A_n g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*) = C_n A_n G_n \Delta_n + C_n A_n R_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*).$$

By Assumption 6(i),

$$C_n A_n G_n = I.$$

Therefore

$$\Delta_n = C_n A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) - C_n A_n R_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*).$$

Taking norms and using the triangle inequality,

$$\|\Delta_n\| \leq \|C_n A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| + \|C_n A_n R_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\|. \quad (37)$$

We first control the Taylor residual. From (36),

$$C_n A_n R_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) = \int_0^1 C_n A_n [J_n(\alpha_n^* + t\Delta_n) - G_n] \Delta_n dt.$$

Since $\alpha_n^* + t\Delta_n \in \mathcal{N}_n(r_n)$ for all $t \in [0, 1]$, Assumption 6(v) implies

$$\|C_n A_n R_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| \leq \rho \|\Delta_n\|.$$

Using this bound in (37),

$$\|\Delta_n\| \leq \|C_n A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| + \rho \|\Delta_n\|.$$

Since $\rho < 1$,

$$\|\Delta_n\| \leq \frac{1}{1-\rho} \|C_n A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\|.$$

Consequently,

$$\|\Delta_n\|^2 \leq \frac{\|C_n\|^2}{(1-\rho)^2} \|A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\|^2. \quad (38)$$

It remains to control

$$\mathbb{E} \left[\|A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\|^2 \right].$$

By definition,

$$e_n(\alpha, \beta_n^*) = \hat{g}_n(\alpha, \beta_n^*) - g_n(\alpha, \beta_n^*),$$

so

$$g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) = \hat{g}_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) - e_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*).$$

Thus,

$$\begin{aligned} \|A_n g_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| &\leq \|A_n \hat{g}_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| \\ &\quad + \|A_n\| \|e_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\|. \end{aligned}$$

We now bound the first term using the sample criterion. By the uniform equivalence of A_n and $\hat{A}_n$ in Assumption 6(ii),

$$\|A_n \hat{g}_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| \leq K \left\| \hat{A}_n \hat{g}_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) \right\|.$$

By the minimizing property of $\hat{\alpha}_n(\beta_n^*)$ under the sample criterion,

$$\left\| \hat{A}_n \hat{g}_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) \right\| \leq \left\| \hat{A}_n \hat{g}_n(\alpha_n^*, \beta_n^*) \right\|.$$

Using again the uniform equivalence of A_n and $\hat{A}_n$,

$$\left\| \hat{A}_n \hat{g}_n(\alpha_n^*, \beta_n^*) \right\| \leq K \|A_n \hat{g}_n(\alpha_n^*, \beta_n^*)\|.$$

Since

$$g_n(\alpha_n^*, \beta_n^*) = 0,$$

we have

$$\hat{g}_n(\alpha_n^*, \beta_n^*) = e_n(\alpha_n^*, \beta_n^*).$$

Using $\|A_n\| \leq K$, we obtain

$$\|A_n \hat{g}_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*)\| \leq K^3 \|e_n(\alpha_n^*, \beta_n^*)\|,$$

where the constant K may change from line to line but is always finite and independent of n .

For the second term,

$$\begin{aligned} e_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) &= e_n(\alpha_n^*, \beta_n^*) \\ &\quad + [e_n(\hat{\alpha}_n(\beta_n^*), \beta_n^*) - e_n(\alpha_n^*, \beta_n^*)]. \end{aligned}$$

Since

$$\hat{\alpha}_n(\beta_n^*) \in \mathcal{N}_n(r_n) \quad \text{and} \quad \mathcal{N}_n(r_n) \subseteq \mathcal{N}_n(\bar{r}),$$

we have

$$\begin{aligned} \|e_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*)\| &\leq \|e_n(\alpha_n^*, \beta_n^*)\| \\ &\quad + \sup_{\alpha \in \mathcal{N}_n(\bar{r})} \|e_n(\alpha, \beta_n^*) - e_n(\alpha_n^*, \beta_n^*)\|. \end{aligned}$$

Combining the previous inequalities and using $\|A_n\| \leq K$, there exists a finite constant $K_1 < \infty$, independent of n , such that

$$\begin{aligned} \|A_n g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*)\| &\leq K_1 \|e_n(\alpha_n^*, \beta_n^*)\| \\ &\quad + K_1 \sup_{\alpha \in \mathcal{N}_n(\bar{r})} \|e_n(\alpha, \beta_n^*) - e_n(\alpha_n^*, \beta_n^*)\|. \end{aligned}$$

Hence,

$$\begin{aligned} \|A_n g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*)\|^2 &\leq 2K_1^2 \|e_n(\alpha_n^*, \beta_n^*)\|^2 \\ &\quad + 2K_1^2 \sup_{\alpha \in \mathcal{N}_n(\bar{r})} \|e_n(\alpha, \beta_n^*) - e_n(\alpha_n^*, \beta_n^*)\|^2. \end{aligned}$$

Taking expectations and using Assumption 6(iv),

$$\begin{aligned} \mathbb{E} \left[\|A_n g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*)\|^2 \right] &\leq 2K_1^2 \mathbb{E} \left[\|e_n(\alpha_n^*, \beta_n^*)\|^2 \right] \\ &\quad + 2K_1^2 \mathbb{E} \left[\sup_{\alpha \in \mathcal{N}_n(\bar{r})} \|e_n(\alpha, \beta_n^*) - e_n(\alpha_n^*, \beta_n^*)\|^2 \right] \\ &= O(q_n^2). \end{aligned}$$

Using (38), we obtain

$$\begin{aligned} \mathbb{E} \left[\|\widehat{\alpha}_n(\beta_n^*) - \alpha_n^*\|^2 \right] &= \mathbb{E} \left[\|\Delta_n\|^2 \right] \\ &\leq \frac{\|C_n\|^2}{(1-\rho)^2} \mathbb{E} \left[\|A_n g_n(\widehat{\alpha}_n(\beta_n^*), \beta_n^*)\|^2 \right] \\ &= O(\|C_n\|^2 q_n^2). \end{aligned}$$

Finally, if $\|C_n\| q_n = o(1)$, then

$$\mathbb{E} \left[\|\widehat{\alpha}_n(\beta_n^*) - \alpha_n^*\|^2 \right] = O(\|C_n\|^2 q_n^2) = o(1).$$

This completes the proof. □

PROOF OF PROPOSITION 2. Let

$$\Delta_{\alpha,n} := \widehat{\alpha}_n(\beta_n^*) - \alpha_n^*.$$

By Assumption 6(iii),

$$\widehat{\alpha}_n(\beta_n^*) \in \mathcal{N}_n(r_n)$$

with probability one for all sufficiently large n . Since $r_n \leq \bar{r}$, we also have

$$\widehat{\alpha}_n(\beta_n^*) \in \mathcal{N}_n(\bar{r})$$

with probability one for all sufficiently large n . Also,

$$\beta_n^* \in \mathcal{B}_n(\bar{s}).$$

Using the mean-value integral form for the map

$$\alpha \mapsto \Gamma(\alpha, \beta_n^*),$$

we obtain

$$\begin{aligned} \widehat{\gamma}_n - \gamma_n^* &= \Gamma(\widehat{\alpha}_n(\beta_n^*), \beta_n^*) - \Gamma(\alpha_n^*, \beta_n^*) \\ &= \int_0^1 \Gamma_{\alpha,n}(\alpha_n^* + t\Delta_{\alpha,n}, \beta_n^*) \Delta_{\alpha,n} dt, \end{aligned}$$

where

$$\Gamma_{\alpha,n}(\alpha, \beta) := \frac{\partial \Gamma(\alpha, \beta)}{\partial \alpha'}.$$

Because $\mathcal{N}_n(\bar{r})$ is convex, and because

$$\alpha_n^*, \widehat{\alpha}_n(\beta_n^*) \in \mathcal{N}_n(\bar{r}),$$

the point

$$(\alpha_n^* + t\Delta_{\alpha,n}, \beta_n^*)$$

belongs to

$$\mathcal{N}_n(\bar{r}) \times \mathcal{B}_n(\bar{s})$$

for all $t \in [0, 1]$. Therefore, by Assumption 6(vi),

$$\begin{aligned}\|\widehat{\gamma}_n - \gamma_n^*\| &\leq \int_0^1 \|\Gamma_{\alpha,n}(\alpha_n^* + t\Delta_{\alpha,n}, \beta_n^*)\| \|\Delta_{\alpha,n}\| dt \\ &\leq L_\Gamma \|\Delta_{\alpha,n}\|.\end{aligned}$$

Squaring gives

$$\|\widehat{\gamma}_n - \gamma_n^*\|^2 \leq L_\Gamma^2 \|\Delta_{\alpha,n}\|^2.$$

Taking expectations,

$$\mathbb{E} [\|\widehat{\gamma}_n - \gamma_n^*\|^2] \leq L_\Gamma^2 \mathbb{E} [\|\widehat{\alpha}_n(\beta_n^*) - \alpha_n^*\|^2].$$

By Proposition 1,

$$\mathbb{E} [\|\widehat{\alpha}_n(\beta_n^*) - \alpha_n^*\|^2] = O(\|C_n\|^2 q_n^2).$$

Hence

$$\mathbb{E} [\|\widehat{\gamma}_n - \gamma_n^*\|^2] = O(\|C_n\|^2 q_n^2).$$

Finally, for any random vector Y and any nonrandom vector y ,

$$E\|Y - y\|^2 = \text{tr}\{\text{Var}(Y)\} + \|E[Y] - y\|^2.$$

Applying this identity with

$$Y = \widehat{\gamma}_n, \quad y = \gamma_n^*,$$

we obtain

$$\text{tr}\{\text{Var}(\widehat{\gamma}_n)\} \leq \mathbb{E} [\|\widehat{\gamma}_n - \gamma_n^*\|^2].$$

Therefore

$$\text{tr}\{\text{Var}(\widehat{\gamma}_n)\} = O(\|C_n\|^2 q_n^2).$$

If $\|C_n\|q_n = o(1)$, then

$$\mathbb{E} [\|\widehat{\gamma}_n - \gamma_n^*\|^2] = O(\|C_n\|^2 q_n^2) = o(1),$$

and

$$\text{tr}\{\text{Var}(\widehat{\gamma}_n)\} = O(\|C_n\|^2 q_n^2) = o(1).$$

Thus, the variance of the object of interest vanishes as $n \rightarrow \infty$. This completes the proof. $\square$

LEMMA 10 (Consistency of the sensitivity statistic). *Suppose Assumptions 1, 2, and 7 hold. Then, for every $S \in \mathcal{S}_{\text{AD}}(\eta^*)$,*

$$\widehat{K}_S(\widehat{\eta}) \xrightarrow{p} K_S(\eta^*).$$

Moreover, since $\mathcal{S}_{\text{AD}}(\eta^*)$ is finite,

$$\max_{S \in \mathcal{S}_{\text{AD}}(\eta^*)} |\widehat{K}_S(\widehat{\eta}) - K_S(\eta^*)| \xrightarrow{p} 0.$$

PROOF. Fix $S \in \mathcal{S}_{\text{AD}}(\eta^*)$. Write

$$G_{\eta,S}(\eta^*) := \frac{\partial g(\eta^*; S)}{\partial \eta'}, \quad \widehat{G}_{\eta,S}(\widehat{\eta}) := \frac{\partial \widehat{g}(\widehat{\eta}; S)}{\partial \eta'}.$$

By Assumption 7(i), $\widehat{\eta} \xrightarrow{p} \eta^*$. By Assumption 7(iii) and continuity of the population Jacobian,

$$\begin{aligned} \|\widehat{G}_{\eta,S}(\widehat{\eta}) - G_{\eta,S}(\eta^*)\| &\leq \sup_{\eta \in N} \left\| \frac{\partial \widehat{g}(\eta; S)}{\partial \eta'} - \frac{\partial g(\eta; S)}{\partial \eta'} \right\| \\ &\quad + \|G_{\eta,S}(\widehat{\eta}) - G_{\eta,S}(\eta^*)\| \\ &\xrightarrow{p} 0. \end{aligned}$$

Therefore the partition-specific submatrices satisfy

$$\widehat{G}_{\alpha,S} \xrightarrow{p} G_{\alpha,S}, \quad \widehat{G}_{\beta,S} \xrightarrow{p} G_{\beta,S}.$$

Since $S \in \mathcal{S}_{\text{AD}}(\eta^*)$, the conditional local identification condition gives

$$\text{rank}(G_{\alpha,S}) = |S|.$$

Because W is positive definite,

$$M_S := G'_{\alpha,S} W G_{\alpha,S}$$

is nonsingular. Hence, with probability approaching one,

$$\widehat{M}_S := \widehat{G}'_{\alpha,S} \widehat{W} \widehat{G}_{\alpha,S}$$

is nonsingular. By the continuous mapping theorem and Assumption 7(iv),

$$\widehat{D}_{\alpha\beta,S} := - \left(\widehat{G}'_{\alpha,S} \widehat{W} \widehat{G}_{\alpha,S} \right)^{-1} \widehat{G}'_{\alpha,S} \widehat{W} \widehat{G}_{\beta,S} \xrightarrow{p} D_{\alpha\beta,S},$$

where

$$D_{\alpha\beta,S} := - \left(G'_{\alpha,S} W G_{\alpha,S} \right)^{-1} G'_{\alpha,S} W G_{\beta,S}.$$

By Assumption 1, Γ is continuously differentiable. Thus

$$\widehat{\Gamma}_{\alpha,S}(\widehat{\eta}) \xrightarrow{p} \Gamma_{\alpha,S}(\eta^*), \quad \widehat{\Gamma}_{\beta,S}(\widehat{\eta}) \xrightarrow{p} \Gamma_{\beta,S}(\eta^*).$$

Therefore

$$\widehat{D}_{\gamma\beta,S} := \widehat{\Gamma}_{\alpha,S}(\widehat{\eta}) \widehat{D}_{\alpha\beta,S} + \widehat{\Gamma}_{\beta,S}(\widehat{\eta}) \xrightarrow{p} D_{\gamma\beta,S},$$

where

$$D_{\gamma\beta,S} := \Gamma_{\alpha,S}(\eta^*) D_{\alpha\beta,S} + \Gamma_{\beta,S}(\eta^*) = \frac{\partial \gamma(\beta^*; S)}{\partial \beta'}.$$

Finally, since the spectral norm is continuous and Assumption 7(iv),

$$\widehat{K}_S(\widehat{\eta}) = \sqrt{|S^c|} \|\widehat{D}_{\gamma\beta,S} \Sigma_S\|_2 \xrightarrow{p} \sqrt{|S^c|} \|D_{\gamma\beta,S} \Sigma_S\|_2 = K_S(\eta^*).$$

This proves pointwise convergence. Since $\mathcal{S}_{\text{AD}}(\eta^*)$ is finite, pointwise convergence implies uniform convergence over $\mathcal{S}_{\text{AD}}(\eta^*)$. $\square$

PROOF OF THEOREM 7. Let

$$\mathcal{S}_0 := \mathcal{S}_{\text{AD}}(\eta^*), \quad S^* := S^*(\eta^*) \in \arg \min_{S \in \mathcal{S}_0} K_S(\eta^*).$$

By Assumption 4, S^* is the unique minimizer. If $\mathcal{S}_0 = \{S^*\}$, then the result follows directly from the admissible-set consistency condition in Assumption 7(ii). Hence suppose $\mathcal{S}_0 \setminus \{S^*\} \neq \emptyset$, and define the strict gap

$$\Delta := \min_{S \in \mathcal{S}_0, S \neq S^*} \{K_S(\eta^*) - K_{S^*}(\eta^*)\}.$$

By uniqueness, $\Delta > 0$.

Define the events

$$A_n := \{\widehat{\mathcal{S}}_{\text{AD}}(\widehat{\eta}) = \mathcal{S}_0\},$$

and

$$B_n := \left\{ \max_{S \in \mathcal{S}_0} |\widehat{K}_S(\widehat{\eta}) - K_S(\eta^*)| < \frac{\Delta}{3} \right\}.$$

By Assumption 7(ii),

$$\Pr(A_n) \rightarrow 1.$$

By Lemma 10,

$$\Pr(B_n) \rightarrow 1.$$

Therefore

$$\Pr(A_n \cap B_n) \rightarrow 1.$$

On $A_n \cap B_n$, the estimated admissible set equals the population admissible set. Moreover, for every $S \in \mathcal{S}_0$ with $S \neq S^*$,

$$\begin{aligned} \widehat{K}_S(\widehat{\eta}) - \widehat{K}_{S^*}(\widehat{\eta}) &= [K_S(\eta^*) - K_{S^*}(\eta^*)] \\ &\quad + [\widehat{K}_S(\widehat{\eta}) - K_S(\eta^*)] - [\widehat{K}_{S^*}(\widehat{\eta}) - K_{S^*}(\eta^*)] \\ &> \Delta - \frac{\Delta}{3} - \frac{\Delta}{3} = \frac{\Delta}{3} > 0. \end{aligned}$$

Thus, on $A_n \cap B_n$, S^* is the unique minimizer of $\widehat{K}_S(\widehat{\eta})$ over $\widehat{\mathcal{S}}_{\text{AD}}(\widehat{\eta})$. Hence

$$\widehat{S}^*(\widehat{\eta}) = S^*(\eta^*)$$

on $A_n \cap B_n$. Therefore,

$$\Pr\{\widehat{S}^*(\widehat{\eta}) \neq S^*(\eta^*)\} \leq \Pr(A_n^c) + \Pr(B_n^c) \rightarrow 0.$$

This proves the consistency of the estimated least-sensitive partition. $\square$

PROOF OF THEOREM 8. Define the sample first-order condition map

$$\widehat{F}(\alpha, \beta; S) := \widehat{G}_\alpha(\alpha, \beta; S)' \widehat{W} \widehat{g}(\alpha, \beta; S).$$

Let

$$\tilde{\eta}_S := (\hat{\alpha}(\hat{\beta}_S; S), \hat{\beta}_S).$$

By Assumption 8,

$$\tilde{\eta}_S \xrightarrow{p} \eta^*.$$

The sample first-order condition is

$$\hat{F}(\tilde{\eta}_S; S) = 0.$$

Let

$$\hat{F}_\alpha(\eta; S) := \frac{\partial \hat{F}(\eta; S)}{\partial \alpha'}, \quad \hat{F}_\beta(\eta; S) := \frac{\partial \hat{F}(\eta; S)}{\partial \beta'}.$$

By the product rule, evaluated at $\tilde{\eta}_S$,

$$\hat{F}_\alpha = \hat{G}'_\alpha \widehat{W} \hat{G}_\alpha + \hat{R}_{\alpha,n},$$

where $\hat{R}_\alpha$ collects the terms involving second derivatives of $\hat{g}$ multiplied by $\widehat{W}\hat{g}$. Similarly,

$$\hat{F}_\beta = \hat{G}'_\alpha \widehat{W} \hat{G}_\beta + \hat{R}_\beta,$$

where $\hat{R}_\beta$ collects the mixed second-derivative terms multiplied by $\widehat{W}\hat{g}$. All matrices in these two displays are evaluated at $\tilde{\eta}_S$.

Because second derivatives are locally bounded with probability approaching one, $\widehat{W} = O_p(1)$, and

$$\hat{g}(\tilde{\eta}_S; S) \xrightarrow{p} g(\eta^*; S) = 0,$$

we have

$$\hat{R}_\alpha = o_p(1), \quad \hat{R}_\beta = o_p(1).$$

Therefore,

$$\hat{F}_\alpha(\tilde{\eta}_S; S) = \hat{G}_\alpha(\tilde{\eta}_S; S)' \widehat{W} \hat{G}_\alpha(\tilde{\eta}_S; S) + o_p(1),$$

and

$$\widehat{F}_\beta(\tilde{\eta}_S; S) = \widehat{G}_\alpha(\tilde{\eta}_S; S)' \widehat{W} \widehat{G}_\beta(\tilde{\eta}_S; S) + o_p(1).$$

Uniform convergence of the Jacobians, consistency of $\tilde{\eta}_S$, and $\widehat{W} \xrightarrow{p} W$ imply

$$\widehat{F}_\alpha(\tilde{\eta}_S; S) \xrightarrow{p} G'_{\alpha,S} W G_{\alpha,S},$$

and

$$\widehat{F}_\beta(\tilde{\eta}_S; S) \xrightarrow{p} G'_{\alpha,S} W G_{\beta,S}.$$

Since $G_{\alpha,S}$ has full column rank and W is positive definite, $G'_{\alpha,S} W G_{\alpha,S}$ is non-singular. Hence $\widehat{F}_\alpha(\tilde{\eta}_S; S)$ is nonsingular with probability approaching one. The sample implicit function theorem gives

$$\left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\widehat{\beta}_S} = - \left[\widehat{F}_\alpha(\tilde{\eta}_S; S) \right]^{-1} \widehat{F}_\beta(\tilde{\eta}_S; S).$$

By the continuous mapping theorem,

$$\left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\widehat{\beta}_S} \xrightarrow{p} -(G'_{\alpha,S} W G_{\alpha,S})^{-1} G'_{\alpha,S} W G_{\beta,S} = D_{\alpha\beta,S}.$$

Now consider the direct derivative used in the algorithm. Since $\widehat{\eta}_n \xrightarrow{p} \eta^*$, the same uniform convergence and continuous mapping argument implies

$$\widehat{D}_{\alpha\beta}(S) \xrightarrow{p} D_{\alpha\beta,S}.$$

Consequently,

$$\left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\widehat{\beta}_S} - \widehat{D}_{\alpha\beta}(S) = o_p(1).$$

Finally, applying the population implicit function theorem to

$$F(\alpha, \beta; S) := G_\alpha(\alpha, \beta; S)' W g(\alpha, \beta; S)$$

at (α^*, β^*) gives

$$\frac{\partial \alpha(\beta^*; S)}{\partial \beta'} = -(G'_{\alpha,S} W G_{\alpha,S})^{-1} G'_{\alpha,S} W G_{\beta,S} = D_{\alpha\beta,S}.$$

The second-derivative terms in the derivative of the population first-order condition vanish because $g(\eta^*; S) = 0$. This completes the proof. $\square$

PROOF OF COROLLARY 1. By the chain rule,

$$\left. \frac{\partial \hat{\gamma}(\beta; S)}{\partial \beta'} \right|_{\beta=\hat{\beta}_S} = \Gamma_\alpha(\tilde{\eta}_S; S) \left. \frac{\partial \hat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\hat{\beta}_S} + \Gamma_\beta(\tilde{\eta}_S; S).$$

Theorem 8 gives the convergence of the derivative of the re-estimation map. Continuity of Γ_α and Γ_β , together with

$$\tilde{\eta}_S \xrightarrow{p} \eta^*, \quad \hat{\eta} \xrightarrow{p} \eta^*,$$

gives

$$\left. \frac{\partial \hat{\gamma}(\beta; S)}{\partial \beta'} \right|_{\beta=\hat{\beta}_S} = D_{\gamma\beta, S} + o_p(1).$$

The same continuity and continuous-mapping argument applied to the direct derivative gives

$$\hat{D}_{\gamma\beta}(S) = D_{\gamma\beta, S} + o_p(1).$$

Therefore

$$\left. \frac{\partial \hat{\gamma}(\beta; S)}{\partial \beta'} \right|_{\beta=\hat{\beta}_S} = \hat{D}_{\gamma\beta}(S) + o_p(1),$$

and the proof is complete. $\square$

APPENDIX D: [

Main Tables: Nakamura and Steinsson Application and Simulation]Main Tables
[Nakamura and Steinsson \(2018\)](#) Application and Simulation

Table 8.: **Set $\hat{S}_{AD}(\hat{\eta})$ and Sensitivity Statistic $\hat{K}(\hat{\beta}, S)$**

<i>Name</i>	<i>Estimated</i>	<i>Calibrated</i>	$\hat{K}(\hat{\beta}; S)$	5% <i>bounds</i>
<i>Original</i>	$(\rho_1, \rho_2, \gamma, \kappa\zeta)$	$(\rho, \delta, \omega, \theta, \sigma, b)$	1.82	[0.58, 0.768]
S^*	$(\delta, \sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \kappa\zeta)$	0.06	[0.67, 0.68]
S_2	$(\rho, \delta, \rho_1, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \kappa\zeta)$	0.32	[0.66, 0.69]
S_3	$(\rho, \rho_1, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \sigma, \kappa\zeta)$	0.36	[0.66, 0.70]
S_4	$(\rho, \delta, \rho_2, \gamma, b)$	$(\omega, \theta, \sigma, \rho_1, \kappa\zeta)$	0.38	[0.66, 0.70]
S_5	$(\sigma, \rho_1, \rho_2, \gamma, b)$	$(\rho, \delta, \omega, \theta, \kappa\zeta)$	0.53	[0.65, 0.70]
S_6	$(\delta, \rho_1, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \sigma, \kappa\zeta)$	0.61	[0.65, 0.71]
S_7	$(\rho, \delta, \rho_1, \gamma, b)$	$(\omega, \theta, \sigma, \rho_2, \kappa\zeta)$	0.61	[0.65, 0.71]
S_8	$(\rho_1, \rho_2, \gamma, b)$	$(\rho, \delta, \omega, \theta, \sigma, \kappa\zeta)$	0.78	[0.64, 0.72]
S_9	$(\rho, \rho_1, \gamma, b)$	$(\delta, \omega, \theta, \sigma, \rho_2, \kappa\zeta)$	0.79	[0.64, 0.72]
S_{10}	$(\delta, \rho_1, \gamma, b)$	$(\rho, \omega, \theta, \sigma, \rho_2, \kappa\zeta)$	0.84	[0.64, 0.72]
S_{11}	(ρ_1, γ, b)	$(\rho, \delta, \omega, \theta, \sigma, \rho_2, \kappa\zeta)$	0.92	[0.63, 0.72]
S_{12}	$(\rho, \delta, \sigma, \rho_1, \rho_2, \gamma)$	$(\omega, \theta, \kappa\zeta, b)$	1.05	[0.63, 0.73]
S_{13}	$(\rho, \sigma, \rho_1, \rho_2, \gamma)$	$(\delta, \omega, \theta, \kappa\zeta, b)$	1.12	[0.62, 0.73]
S_{14}	$(\rho, \sigma, \rho_1, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \kappa\zeta)$	1.19	[0.62, 0.74]
S_{15}	$(\rho, \delta, \sigma, \rho_1, \gamma)$	$(\omega, \theta, \rho_2, \kappa\zeta, b)$	1.22	[0.62, 0.74]
S_{16}	$(\rho, \delta, \sigma, \rho_2, \gamma)$	$(\omega, \theta, \rho_1, \kappa\zeta, b)$	1.23	[0.62, 0.74]
S_{17}	$(\sigma, \rho_1, \gamma, b)$	$(\rho, \delta, \omega, \theta, \rho_2, \kappa\zeta)$	1.25	[0.61, 0.74]

Table 8 (continued)

<i>Name</i>	<i>Estimated Parameters</i>	<i>Fixed Parameters</i>	$\hat{K}(\hat{\beta}; S)$	<i>5% bounds</i>
S_{18}	$(\delta, \sigma, \rho_1, \gamma, b)$	$(\rho, \omega, \theta, \rho_2, \kappa\zeta)$	1.26	[0.61, 0.74]
S_{19}	$(\delta, \sigma, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \rho_1, \kappa\zeta)$	1.34	[0.61, 0.74]
S_{20}	$(\rho, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \sigma, \rho_1, \kappa\zeta)$	1.53	[0.60, 0.75]
S_{21}	$(\rho, \sigma, \rho_2, \gamma)$	$(\delta, \omega, \theta, \rho_1, \kappa\zeta, b)$	1.53	[0.60, 0.75]
S_{22}	$(\rho, \sigma, \rho_1, \gamma)$	$(\delta, \omega, \theta, \rho_2, \kappa\zeta, b)$	1.59	[0.60, 0.76]
S_{23}	$(\delta, \sigma, \rho_1, \rho_2, \gamma)$	$(\rho, \omega, \theta, \kappa\zeta, b)$	1.63	[0.60, 0.76]
S_{24}	$(\rho, \sigma, \rho_1, \gamma, b)$	$(\delta, \omega, \theta, \rho_2, \kappa\zeta)$	1.73	[0.59, 0.76]
S_{25}	$(\delta, \rho_1, \rho_2, \gamma)$	$(\rho, \omega, \theta, \sigma, \kappa\zeta, b)$	1.82	[0.59, 0.77]
S_{26}	$(\delta, \sigma, \rho_1, \gamma)$	$(\rho, \omega, \theta, \rho_2, \kappa\zeta, b)$	1.86	[0.58, 0.77]
S_{27}	$(\rho, \delta, \rho_1, \rho_2, \gamma)$	$(\omega, \theta, \sigma, \kappa\zeta, b)$	1.93	[0.58, 0.77]
S_{28}	$(\rho, \delta, \rho_1, \gamma)$	$(\omega, \theta, \sigma, \rho_2, \kappa\zeta, b)$	1.94	[0.58, 0.77]
S_{29}	$(\sigma, \rho_2, \gamma, b)$	$(\rho, \delta, \omega, \theta, \rho_1, \kappa\zeta)$	1.96	[0.58, 0.78]
S_{30}	(δ, ρ_1, γ)	$(\rho, \omega, \theta, \sigma, \rho_2, \kappa\zeta, b)$	1.96	[0.58, 0.78]
S_{31}	$(\sigma, \rho_1, \rho_2, \gamma)$	$(\rho, \delta, \omega, \theta, \kappa\zeta, b)$	1.98	[0.58, 0.78]
S_{32}	(ρ_1, ρ_2, γ)	$(\rho, \delta, \omega, \theta, \sigma, \kappa\zeta, b)$	1.99	[0.58, 0.78]
S_{33}	$(\rho, \rho_1, \rho_2, \gamma)$	$(\delta, \omega, \theta, \sigma, \kappa\zeta, b)$	2.07	[0.57, 0.78]
S_{34}	(σ, ρ_1, γ)	$(\rho, \delta, \omega, \theta, \rho_2, \kappa\zeta, b)$	2.09	[0.57, 0.78]
S_{35}	(ρ_1, γ)	$(\rho, \delta, \omega, \theta, \sigma, \rho_2, \kappa\zeta, b)$	2.13	[0.57, 0.78]

Table 8 (continued)

<i>Name</i>	<i>Estimated Parameters</i>	<i>Fixed Parameters</i>	$\hat{K}(\hat{\beta}; S)$	<i>5% bounds</i>
S_{36}	(ρ, ρ_1, γ)	$(\delta, \omega, \theta, \sigma, \rho_2, \kappa\zeta, b)$	2.16	[0.57, 0.79]
S_{37}	$(\rho, \delta, \rho_2, \gamma)$	$(\omega, \theta, \sigma, \rho_1, \kappa\zeta, b)$	2.39	[0.56, 0.80]
S_{38}	(ρ, ρ_2, γ)	$(\delta, \omega, \theta, \sigma, \rho_1, \kappa\zeta, b)$	2.89	[0.53, 0.82]
S_{39}	$(\rho, \sigma, \rho_2, \gamma, b)$	$(\delta, \omega, \theta, \rho_1, \kappa\zeta)$	3.70	[0.49, 0.86]
S_{40}	$(\delta, \sigma, \rho_2, \gamma)$	$(\rho, \omega, \theta, \rho_1, \kappa\zeta, b)$	3.71	[0.49, 0.86]
S_{41}	(σ, ρ_2, γ)	$(\rho, \delta, \omega, \theta, \rho_1, \kappa\zeta, b)$	4.46	[0.45, 0.90]
S_{42}	$(\delta, \rho_2, \gamma, b)$	$(\rho, \omega, \theta, \sigma, \rho_1, \kappa\zeta)$	4.85	[0.43, 0.92]
S_{43}	(δ, ρ_2, γ)	$(\rho, \omega, \theta, \sigma, \rho_1, \kappa\zeta, b)$	4.94	[0.43, 0.92]
S_{44}	(ρ_2, γ)	$(\rho, \delta, \omega, \theta, \sigma, \rho_1, \kappa\zeta, b)$	5.56	[0.40, 0.96]
S_{45}	(ρ_2, γ, b)	$(\rho, \delta, \omega, \theta, \sigma, \rho_1, \kappa\zeta)$	5.57	[0.40, 0.96]
S_{46}	$(\rho, \delta, \sigma, \gamma)$	$(\omega, \theta, \rho_1, \rho_2, \kappa\zeta, b)$	9.25	[0.22, 1.14]
S_{47}	$(\rho, \delta, \gamma, b)$	$(\omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta)$	9.74	[0.19, 1.16]
S_{48}	(δ, σ, γ)	$(\rho, \omega, \theta, \rho_1, \rho_2, \kappa\zeta, b)$	9.83	[0.19, 1.17]
S_{49}	(ρ, δ, γ)	$(\omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta, b)$	9.95	[0.18, 1.17]
S_{50}	(ρ, σ, γ)	$(\delta, \omega, \theta, \rho_1, \rho_2, \kappa\zeta, b)$	10.00	[0.18, 1.18]
S_{51}	(δ, γ, b)	$(\rho, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta)$	10.15	[0.17, 1.19]
S_{52}	$(\delta, \sigma, \gamma, b)$	$(\rho, \omega, \theta, \rho_1, \rho_2, \kappa\zeta)$	10.23	[0.17, 1.19]
S_{53}	$(\rho, \sigma, \gamma, b)$	$(\delta, \omega, \theta, \rho_1, \rho_2, \kappa\zeta)$	10.43	[0.16, 1.20]

Table 8 (continued)

<i>Name</i>	<i>Estimated Parameters</i>	<i>Fixed Parameters</i>	$\hat{K}(\hat{\beta}; S)$	<i>5% bounds</i>
S_{54}	(δ, γ)	$(\rho, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta, b)$	10.43	[0.16, 1.20]
S_{55}	(σ, γ)	$(\rho, \delta, \omega, \theta, \rho_1, \rho_2, \kappa\zeta, b)$	10.52	[0.15, 1.20]
S_{56}	(ρ, γ, b)	$(\delta, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta)$	10.55	[0.15, 1.20]
S_{57}	(ρ, γ)	$(\delta, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta, b)$	10.65	[0.15, 1.21]
S_{58}	(γ, b)	$(\rho, \delta, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta)$	10.88	[0.13, 1.22]
S_{59}	(σ, γ, b)	$(\rho, \delta, \omega, \theta, \rho_1, \rho_2, \kappa\zeta)$	11.02	[0.13, 1.23]
S_{60}	(γ)	$(\rho, \delta, \omega, \theta, \sigma, \rho_1, \rho_2, \kappa\zeta, b)$	11.07	[0.12, 1.23]

Table 9.: **Contribution of Fixed Parameters to Sensitivity Statistic**

<i>Name</i>	ρ	δ	ω	θ	σ	ρ_1	ρ_2	$\kappa\zeta$	γ	b
<i>Original</i>	0.0%	0.0%	0.0%	0.0%	0.0%	-	-	-	-	100.0%
S^*	99.6%	-	0.4%	0.0%	-	-	-	0.0%	-	-
S_2	-	-	0.0%	0.0%	100.0%	-	-	0.0%	-	-
S_3	-	22.9%	2.0%	4.3%	70.7%	-	-	0.2%	-	-
S_4	-	-	0.0%	0.0%	87.6%	12.4%	-	0.0%	-	-
S_5	0.6%	77.7%	6.6%	14.6%	-	-	-	0.5%	-	-
S_6	0.6%	-	0.0%	0.0%	99.4%	-	-	0.0%	-	-

Table 9 (continued)

<i>Name</i>	ρ	δ	ω	θ	σ	ρ_1	ρ_2	$\kappa\zeta$	γ	b
S_7	-	-	0.0%	0.0%	33.8%	-	66.2%	0.0%	-	-
S_8	0.7%	2.6%	0.2%	0.5%	96.1%	-	-	0.0%	-	-
S_9	-	1.2%	0.1%	0.2%	27.8%	-	70.6%	0.0%	-	-
S_{10}	0.8%	-	0.0%	0.0%	86.8%	-	12.4%	0.0%	-	-
S_{11}	0.8%	1.6%	0.1%	0.3%	93.2%	-	4.1%	0.0%	-	-
S_{12}	-	-	0.0%	0.0%	-	-	-	0.0%	-	100.0%
S_{13}	-	0.2%	0.0%	0.0%	-	-	-	0.0%	-	99.7%
S_{14}	-	78.0%	6.8%	14.7%	-	-	-	0.5%	-	-
S_{15}	-	-	0.0%	0.0%	-	-	7.5%	0.0%	-	92.5%
S_{16}	-	-	0.0%	0.0%	-	10.2%	-	0.0%	-	89.8%
S_{17}	0.3%	0.3%	0.0%	0.0%	-	-	99.4%	0.0%	-	-
S_{18}	0.3%	-	0.0%	0.0%	-	-	99.7%	0.0%	-	-
S_{19}	0.2%	-	0.0%	0.0%	-	99.8%	-	0.0%	-	-
S_{20}	-	8.9%	0.7%	1.7%	2.7%	85.9%	-	0.1%	-	-
S_{21}	-	2.9%	0.3%	0.6%	-	52.0%	-	0.0%	-	44.2%
S_{22}	-	1.3%	0.1%	0.3%	-	-	13.6%	0.0%	-	84.7%
S_{23}	0.0%	-	0.0%	0.0%	-	-	-	0.0%	-	100.0%
S_{24}	-	46.2%	3.9%	8.7%	-	-	40.9%	0.3%	-	-

Table 9 (continued)

<i>Name</i>	ρ	δ	ω	θ	σ	ρ_1	ρ_2	$\kappa\zeta$	γ	b
S_{25}	0.0%	-	0.0%	0.0%	0.0%	-	-	0.0%	-	100.0%
S_{26}	0.0%	-	0.0%	0.0%	-	-	0.2%	0.0%	-	99.7%
S_{27}	-	-	0.0%	0.0%	1.4%	-	-	0.0%	-	98.6%
S_{28}	-	-	0.0%	0.0%	1.0%	-	2.1%	0.0%	-	97.0%
S_{29}	0.0%	16.4%	1.4%	3.1%	-	79.0%	-	0.1%	-	-
S_{30}	0.0%	-	0.0%	0.0%	0.1%	-	0.6%	0.0%	-	99.3%
S_{31}	0.0%	0.3%	0.0%	0.1%	-	-	-	0.0%	-	99.6%
S_{32}	0.1%	0.6%	0.0%	0.1%	0.3%	-	-	0.0%	-	98.9%
S_{33}	-	0.0%	0.0%	0.0%	1.4%	-	-	0.0%	-	98.5%
S_{34}	0.0%	0.5%	0.0%	0.1%	-	-	0.4%	0.0%	-	98.9%
S_{35}	0.0%	0.6%	0.0%	0.1%	0.2%	-	0.0%	0.0%	-	98.9%
S_{36}	-	0.5%	0.0%	0.1%	0.6%	-	5.3%	0.0%	-	93.5%
S_{37}	-	-	0.0%	0.0%	1.6%	7.7%	-	0.0%	-	90.7%
S_{38}	-	1.6%	0.1%	0.3%	1.7%	33.9%	-	0.0%	-	62.4%
S_{39}	-	10.1%	0.9%	1.9%	-	87.1%	-	0.1%	-	-
S_{40}	0.0%	-	0.0%	0.0%	-	47.8%	-	0.0%	-	52.2%
S_{41}	0.0%	0.5%	0.0%	0.1%	-	65.2%	-	0.0%	-	34.2%
S_{42}	0.2%	-	0.0%	0.0%	8.8%	91.0%	-	0.0%	-	-

Table 9 (continued)

<i>Name</i>	ρ	δ	ω	θ	σ	ρ_1	ρ_2	$\kappa\zeta$	γ	b
S_{43}	0.1%	-	0.0%	0.0%	3.5%	86.9%	-	0.0%	-	9.5%
S_{44}	0.1%	0.1%	0.0%	0.0%	2.7%	89.0%	-	0.0%	-	8.1%
S_{45}	0.1%	0.1%	0.0%	0.0%	6.7%	93.0%	-	0.0%	-	-
S_{46}	-	-	0.0%	0.0%	-	80.0%	18.1%	0.0%	-	1.9%
S_{47}	-	-	0.0%	0.0%	0.2%	80.2%	19.6%	0.0%	-	-
S_{48}	0.0%	-	0.0%	0.0%	-	80.6%	16.7%	0.0%	-	2.7%
S_{49}	-	-	0.0%	0.0%	0.0%	79.8%	17.9%	0.0%	-	2.3%
S_{50}	-	0.0%	0.0%	0.0%	-	80.1%	18.0%	0.0%	-	1.9%
S_{51}	0.0%	-	0.0%	0.0%	0.6%	84.8%	14.6%	0.0%	-	-
S_{52}	0.0%	-	0.0%	0.0%	-	74.4%	25.6%	0.0%	-	-
S_{53}	-	1.5%	0.1%	0.3%	-	89.5%	8.6%	0.0%	-	-
S_{54}	0.0%	-	0.0%	0.0%	0.0%	82.8%	15.1%	0.0%	-	2.1%
S_{55}	0.0%	0.0%	0.0%	0.0%	-	81.1%	16.3%	0.0%	-	2.6%
S_{56}	-	0.0%	0.0%	0.0%	0.1%	81.0%	18.9%	0.0%	-	-
S_{57}	-	0.0%	0.0%	0.0%	0.0%	80.0%	17.7%	0.0%	-	2.3%
S_{58}	0.0%	0.0%	0.0%	0.0%	0.5%	85.2%	14.2%	0.0%	-	-
S_{59}	0.0%	0.0%	0.0%	0.0%	-	77.4%	22.6%	0.0%	-	-
S_{60}	0.0%	0.0%	0.0%	0.0%	0.0%	82.9%	15.0%	0.0%	-	2.1%

APPENDIX E: MONTE CARLO APPENDIX

The Monte Carlo exercise is implemented as follows. Let η_0 denote the reference structural parameter vector used in the empirical application. For each value of η_0 , we solve the structural model and compute the model-implied vector

$$m(\eta_0) \in \mathbb{R}^{33},$$

whose entries correspond to the same reduced-form responses matched in the CMD estimation.

The simulated observations are draws of this stacked reduced-form moment vector. Specifically, for each Monte Carlo replication and sample size n , we draw

$$m_i = m(\eta_0) + u_i, \quad u_i \sim N(0, \hat{\Omega}), \quad i = 1, \dots, n,$$

where $\hat{\Omega}$ is the covariance matrix computed from the bootstrap distribution of the empirical reduced-form moments. We then compute the sample mean and sample covariance of $\{m_i\}_{i=1}^n$ and use these objects to construct the same moment structure that enters the empirical CMD estimator.

Thus, the Monte Carlo sample size n indexes the precision of the simulated reduced-form moment vector, rather than the number of raw daily or tick observations. We report results for $n = 150$, which is of the same order as the number of FOMC-event observations underlying the empirical reduced-form moments, and for $n = 500$, which provides a larger-sample benchmark in which sampling noise is smaller.

For each partition S and each miscalibration size ε , we compute the partition-specific worst-case perturbation of the fixed parameters. The fixed parameters are held at these miscalibrated values, while the estimated parameters are re-estimated using the simulated CMD moments. Across Monte Carlo replications, this delivers the finite-sample distribution of $\hat{\gamma}$ under each partition and miscalibration size. We summarize these distributions using bias, variance, and mean squared error relative to the reference value γ_0 .

E.1 Robustness to the weak-identification rank threshold

This appendix checks whether the Monte Carlo conclusions depend on the hard-threshold used to estimate the conditional rank of each partition. In the baseline implementation, a partition is kept whenever

$$\hat{r}_S = |S|, \quad \hat{r}_S = \#\{j : \hat{\sigma}_{j,S} > \tau_n\}, \quad \tau_n = \left(\frac{\log n}{n}\right)^{1/2}.$$

To study sensitivity to this implementation choice, we replace the baseline cutoff by

$$\tau_n(a) = \left(\frac{\log n}{n}\right)^a, \quad a \in \{0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}.$$

The scalar a is not a structural parameter. It is the exponent applied to $\log n/n$ in the rank threshold. Since $\log n/n < 1$ in the samples considered here, larger values of a imply a smaller cutoff and therefore a more permissive admissibility rule. For each value of a , I recompute the admissible set, rank admissible partitions by the sensitivity statistic K_S , and repeat the same worst-case Monte Carlo exercise for the top-ranked partitions.

Table 10 reports the results in compressed form. The first panel shows how the top-five sensitivity ranking changes with a . The second panel reports Monte Carlo MSEs only for the distinct partitions that appear in one of those rankings. This avoids repeating identical rows: for $a \in \{0.50, 0.60, 0.70, 0.80\}$, both the number of admissible partitions and the top-five ranking are exactly the same. The main conclusion is stable. The selected partition S^* remains the least-sensitive partition for every threshold exponent. When the cutoff is made more permissive, additional partitions enter the admissible set, but they do not overturn the main finite-sample message: the least-sensitive partition remains unchanged and, when miscalibration is large relative to the sample size, delivers the lowest MSE.

TABLE 10. Robustness to the rank-threshold exponent

Threshold exponent a	$\#\hat{\mathcal{S}}_{AD}(a)$	Top five by K_S	Sensitivities	Main change relative to baseline
$a \in \{0.50, 0.60, 0.70, 0.80\}$	60	S^*, S_2, S_3, S_4, S_5	0.0582, 0.3159, 0.3602, 0.3772, 0.5324	No change: same admissible set size and same top-five ranking.
$a = 0.90$	86	$S^*, S_2, S_3, S_4, S_{0.9}$	0.0582, 0.3159, 0.3602, 0.3772, 0.3860	More permissive threshold admits additional partitions; the first four positions are unchanged.
$a = 1.00$	95	$S^*, S_{1a}, S_{1b}, S_{1c}, S_2$	0.0582, 0.0591, 0.0600, 0.3153, 0.3159	Near-ties appear only under the most permissive threshold, but S^* remains first.

Partition	Estimated parameters	Fixed parameters	K_S	MSE, $n = 150$			MSE, $n = 500$		
				$\varepsilon = 5\%$	$\varepsilon = 10\%$	$\varepsilon = 15\%$	$\varepsilon = 5\%$	$\varepsilon = 10\%$	$\varepsilon = 15\%$
S^*	$\delta, \sigma, \rho_1, \rho_2, \gamma, b$	$\rho, \omega, \theta, \kappa\zeta$	0.0582	4.7	5.5	6.7	1.4	1.8	2.5
S_2	$\rho, \delta, \rho_1, \rho_2, \gamma, b$	$\omega, \theta, \sigma, \kappa\zeta$	0.3159	5.7	8.6	22.4	4.5	7.0	21.0
S_3	$\rho, \rho_1, \rho_2, \gamma, b$	$\delta, \omega, \theta, \sigma, \kappa\zeta$	0.3602	10.7	8.8	12.9	9.6	7.3	11.7
S_4	$\rho, \delta, \rho_2, \gamma, b$	$\omega, \theta, \sigma, \rho_1, \kappa\zeta$	0.3772	21.1	349.3	1492.8	20.4	345.1	1494.3
S_5	$\sigma, \rho_1, \rho_2, \gamma, b$	$\rho, \delta, \omega, \theta, \kappa\zeta$	0.5324	79.7	448.9	602.8	78.1	437.9	605.3
$S_{0.9}$	$\rho, \omega, \rho_2, \gamma, b$	$\delta, \theta, \sigma, \rho_1, \kappa\zeta$	0.3860	56.8	653.2	3449.9	56.9	648.5	3540.0
S_{1a}	$\rho, \delta, \sigma, \rho_1, \rho_2, \gamma, b$	$\omega, \theta, \kappa\zeta$	0.0591	54.1	51.7	52.2	15.7	16.9	17.2
S_{1b}	$\omega, \sigma, \rho_1, \rho_2, \gamma, b$	$\rho, \delta, \theta, \kappa\zeta$	0.0600	4.0	6.7	18.2	1.2	4.1	16.0
S_{1c}	$\rho, \omega, \rho_1, \rho_2, \gamma, b$	$\delta, \theta, \sigma, \kappa\zeta$	0.3153	5.7	9.0	31.3	4.5	7.3	29.3

Notes: The rank-threshold exponent a defines $\tau_n(a) = (\log n/n)^a$. Larger a corresponds to a smaller singular-value cutoff and therefore a more permissive admissibility rule. K_S is the sensitivity statistic used to rank admissible partitions. MSEs are multiplied by 10^4 and rounded to one decimal place. The table reports MSEs, rather than bias and variance separately, to focus on the robustness of the substantive ranking; the worst-case direction is recomputed separately for each partition and each value of ε , so the sign of the bias is not the object of comparison.

APPENDIX: REFERENCES

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